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Mid Sem Assignment - Electrodynamics.

Chapter 1: Vector Analysis.

Theory.

Q1. Describe dot and cross product of 2 vectors?

Dot product of 2 vectors is defined by $\boxed{A \cdot B = AB\cos\theta}$ where θ is the angle they form when placed tail to tail

Cross product of 2 vectors is defined by $\boxed{A \times B = AB\sin\theta\,\hat{n}}$ where $\hat{n}$ is a unit vector pointing perpendicular to the plane of A and B

Q2 Discuss scalar triple product & Vector triple product.

Since the cross product of 2 vectors is itself a vector, it can be dotted or crossed with a third vector to form a triple product

(i) Scalar triple product : $\boxed{A \cdot (B \times C)}$. Geometrically it is the volume of the parallelopided generated by A, B and C, since $|B \times C|$ is the area of the base, and $|A\cos\theta|$ is the altitude

(ii) Vector triple product : $\boxed{A \times (B \times C)}$ The vector triple product can be simplified by the so-called BAC-CAB rule:

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$$

Q3. Define Del operator.

$$\boxed{\nabla = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}}$$

∇ is a vector operator that acts upon T. It is an instruction to differentiate what follows.

Q4. Describe Gradient, Divergence & Curl of any function.

Gradient $\nabla T = \frac{\partial T}{\partial x}\hat{x} + \frac{\partial T}{\partial y}\hat{y} + \frac{\partial T}{\partial z}\hat{z}$ is the gradient of T

The gradient ∇T points in the direction of maximum increase of function T. The magnitude $|\nabla T|$ gives the slope (rate of increase) along this maximal direction.

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Divergence $\nabla \cdot v = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right) \cdot (v_x \hat{x} + v_y \hat{y} + v_z \hat{z})$

$$= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Divergence is a measure of how much the vector v spreads out (diverges) from the point in question

Curl $\nabla \times v = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} = \hat{x}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{y}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{z}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)$

Curl is a measure of how much the vector v swirls around the point in question.

Q5. Discuss the fundamental theorem of gradients, divergence & curl.

Fundamental Theorem of Gradients

Integral (line integral) of derivative (gradient) over some region is given by value of function at the boundaries (a, & b)

$$\int_a^b (\nabla T) \cdot dl = T(b) - T(a)$$

Fundamental Theorem of Divergence

Integral of derivative (divergence) over a region (volume) is given by value of function at the boundaries (surface that bounds volume)

$$\int (\nabla \cdot v) d\tau = \oint_S v \cdot da$$

Fundamental Theorem of Curls

Integral of derivative (curl) over a region (path of surface) is given by value of function at the boundaries (perimeter of patch)

$$\int_S (\nabla \times v) \cdot da = \oint_P v \cdot dl$$

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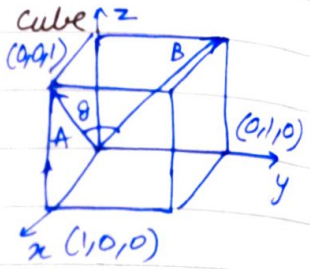
Example

1.1 Let $C = A - B$ and calculate the dot product of C with itself.

$$C \cdot C = (A-B)(A-B) = A \cdot A - A \cdot B - B \cdot A + B \cdot B$$

$$C^2 = A^2 + B^2 - 2AB\cos\theta$$

1.2 Find the angle between the face diagonals of a cube



The face diagonals A and B are

$$A = 1\hat{x} + 0\hat{y} + 1\hat{z} \; ; \; B = 0\hat{x} + 1\hat{y} + 1\hat{z}$$

$$A \cdot B = 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 = 1$$

$$A \cdot B = AB\cos\theta = \sqrt{2}\sqrt{2}\cos\theta = 2\cos\theta$$

$$\cos\theta = 1/2 \quad \text{or} \quad \theta = 60^\circ$$

1.3 Find the gradient of $r = \sqrt{x^2+y^2+z^2}$ (the magnitude of position vector)

$$\nabla r = \frac{\partial r}{\partial x}\hat{x} + \frac{\partial r}{\partial y}\hat{y} + \frac{\partial r}{\partial z}\hat{z}$$

$$= \frac{1}{2}\frac{2x}{\sqrt{x^2+y^2+z^2}}\hat{x} + \frac{1}{2}\frac{2y}{\sqrt{x^2+y^2+z^2}}\hat{y} + \frac{1}{2}\frac{2z}{\sqrt{x^2+y^2+z^2}}\hat{z}$$

$$= \frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2+y^2+z^2}} = \frac{\mathbf{r}}{r} = \hat{r}$$

1.4 Suppose the functions are $v_a = r = x\hat{x} + y\hat{y} + z\hat{z}$, $v_b = \hat{z}$ and $v_c = z\hat{z}$. Calculate their divergences.

$$\nabla \cdot v_a = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

$$\nabla \cdot v_b = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(1) = 0 + 0 + 0 = 0$$

$$\nabla \cdot v_c = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 0 + 0 + 1 = 1$$

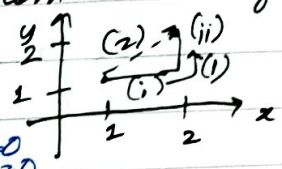
1.5 Suppose the function is $v_a = -y\hat{x} + x\hat{y}$ and $v_b = x\hat{y}$. Calculate their curls

$$\nabla \times v_a = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix} = 2\hat{z}$$

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$$\nabla \times v_b = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x & 0 \end{vmatrix} = \hat{z}$$

1.6 Calculate the line integral of the function $v = y^2\hat{x} + 2x(y+1)\hat{y}$ from the point $a = (1,1,0)$ to the point $b = (2,2,0)$ along the paths (1) and (2) in fig. What is $\oint v \cdot dl$ for the loop that goes from a to b along (1) and returns to a along (2)?



$dl = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Path (1) consist of 2 parts. Along horizontal $dy = 0$, $dz = 0$

(i) $dl = dx\hat{x}$, $y = 1$, $v \cdot dl = y^2 dx = dx$ so $\int v \cdot dl = \int_1^2 dx = 1$

on vertical stretch $dx = dz = 0$ so

(ii) $dl = dy\hat{y}$, $x = 2$, $v \cdot dl = 2x(y+1)dy = 4(y+1)dy$

$$\int v \cdot dl = 4\int_1^2 (y+1)\,dy = 10$$

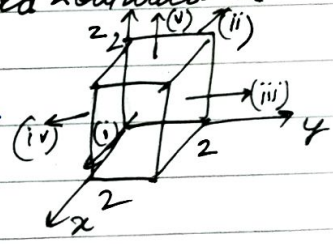
By path (1) then, $\int_a^b v \cdot dl = 1 + 10 = 11$

on path (2) $x = y$, $dx = dy$ and $dz = 0$

$dl = dx\hat{x} + dx\hat{y}$, $v \cdot dl = x^2 dx + 2x(x+1)dx = (3x^2 + 2x)dx$

$$\int_a^b v \cdot dl = \int_1^2 (3x^2 + 2x)\,dx = (x^3 + x^2)\Big|_1^2 = 10$$

1.7 Calculate the surface integral of $v = 2xz\hat{x} + (x+2)\hat{y} + y(z^2-3)\hat{z}$ over 5 sides of cubical box in fig. Let upward & outward be +ve dir^n



Taking sides one at a time

(i) $x = 2$, $da = dy\,dz\,\hat{x}$, $v \cdot da = 2xz\,dy\,dz = 4z\,dy\,dz$

$$\int v \cdot da = 4\int_0^2 dy \int_0^2 z\,dz = 16$$

(ii) $x = 0$, $da = -dy\,dz\,\hat{x}$, $v \cdot da = -2xz\,dy\,dz = 0$

$$\int v \cdot da = 0$$

(iii) $y = 2$, $da = dx\,dz\,\hat{y}$, $v \cdot da = (x+2)\,dx\,dz$

$$\int v \cdot da = \int_0^2 (x+2)\,dx \int_0^2 dz = 12$$

(iv) $y = 0$, $da = -dx\,dz\,\hat{y}$, $v \cdot da = -(x+2)\,dx\,dz$

$$\int v \cdot da = -\int_0^2 (x+2)\,dx \int_0^2 dz = -12$$

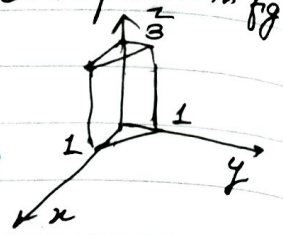
(v) $z = 2$, $da = dx\,dy\,\hat{z}$, $v \cdot da = y(z^2-3)\,dx\,dy = y\,dx\,dy$

$$\int v \cdot da = \int_0^2 dx \int_0^2 y\,dy = 4$$

Total flux is $\int_{\text{surface}} v \cdot da = 16 + 0 + 12 - 12 + 4 = 20$

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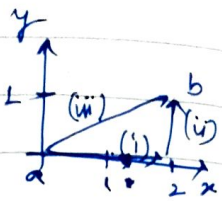
1.8 Calculate the volume integral of $T = xyz^2$ over the prism in fig.



$$\int T\,d\tau = \int_0^3 z^2 \int_0^1 y \int_0^{1-y} x\,dx\,dy\,dz$$

$$= \frac{1}{2}\int_0^3 z^2\,dz \int_0^1 (1-y)^2 y\,dy = \frac{1}{2}\times 9 \times \frac{1}{12} = 3/8$$

1.9 Let $T = xy^2$, and take point a to be the origin $(0,0,0)$ and b to the point $(2,1,0)$. Check fundamental theorem for gradients



$$dl = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$$

$$\nabla T = y^2\hat{x} + 2xy\hat{y}$$

(i) $y = 0$; $dl = dx\,\hat{x}$, $\nabla T \cdot dl = y^2 dx = 0$

$$\int \nabla T \cdot dl = 0$$

(ii) $x = 2$; $dl = dy\,\hat{y}$, $\nabla T \cdot dl = 2xy\,dy = 4y\,dy$

$$\int_{ii} \nabla T \cdot dl = \int_0^1 4y\,dy = 2y^2\Big|_0^1 = 2$$

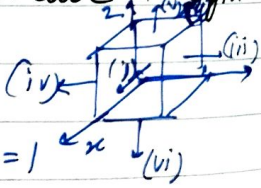
consistent with fundamental theorem

$$T(b) - T(a) = 2 - 0 = 2$$

(iii) $y = \frac{1}{2}x$, $dy = \frac{1}{2}dx$, $\nabla T \cdot dl = y^2 dx + 2xy\,dy = \frac{3}{4}x^2 dx$

$$\int_{iii} \nabla T \cdot dl = \int_0^2 3/4\,x^2 dx = \frac{1}{4}x^3\Big|_0^2 = 2$$

1.10 Check the divergence theorem using the function $v = y^2\hat{x} + (2xy + z^2)\hat{y} + (2yz)\hat{z}$ and a unit cube at origin



$$\nabla \cdot v = 2(x+y)$$

$$\int_v 2(x+y)\,d\tau = 2\int_0^1\int_0^1\int_0^1 (x+y)\,dx\,dy\,dz$$

$$\int_0^1 (x+y)\,dx = \frac{1}{2} + y, \quad \int_0^1 \left(\frac{1}{2}+y\right)dy = 1, \quad \int_0^1 1\,dz = 1$$

$$\int_v \nabla \cdot v\,d\tau = 2$$

(i) $\int v \cdot da = \int_0^1\int_0^1 y^2\,dy\,dz = 1/3$

(ii) $\int v \cdot da = -\int_0^1\int_0^1 y^2\,dy\,dz = -1/3$

(iii) $\int v \cdot da = \int_0^1\int_0^1 (2x + z^2)\,dx\,dz = 4/3$

(iv) $\int v \cdot da = -\int_0^1\int_0^1 z^2\,dx\,dz = -1/3$

(v) $\int v \cdot da = \int_0^1\int_0^1 2y\,dx\,dy = 1$

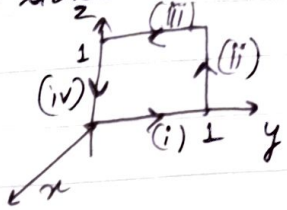
(vi) $\int v \cdot da = -\int_0^1\int_0^1 0\,dx\,dy = 0$

Total flux is: $\oint_S v \cdot da = \frac{1}{3} - \frac{1}{3} + \frac{4}{3} - \frac{1}{3} + 1 + 0 = 2$

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1.11 Suppose $v = (2xz + 3y^2)\hat{y} + (4yz^2)\hat{z}$. Check Stokes' Theorem for the square surface shown in Fig.



$\nabla \times v = (4z^2 - 2x)\hat{x} + 2z\hat{z}$ and $da = dy\,dz\,\hat{x}$

$\int (\nabla \times v)\, da = \int_0^1 \int_0^1 4z^2\, dy\, dz = 4/3$

Line integral

(i) $x=0$, $z=0$, $v \cdot dl = 3y^2 dy$, $\int v \cdot dl = \int_0^1 3y^2 dy = 1$

(ii) $x=0$, $y=1$, $v \cdot dl = 4z^2 dz$, $\int v \cdot dl = \int_0^1 4z^2 = dz = 4/3$

(iii) $x=0$, $z=1$, $v \cdot dl = 3y^2 dy$, $\int v \cdot dl = \int_1^0 3y^2 dy = -1$

(iv) $x=0$, $y=0$, $v \cdot dl = 0$, $\int v \cdot dl = \int_1^0 0\, dz = 0$

so $\oint v \cdot dl = 1 + 4/3 - 1 + 0 = 4/3$

1.12 Evaluate the integral

$$\int_0^\infty x e^{-x} dx$$

$$e^{-x} = \frac{d}{dx}(-e^{-x})$$

$f(x) = x$, $g(x) = -e^{-x}$ and $df/dx = 1$

$$\int_0^\infty x e^{-x} dx = \int_0^\infty e^{-x} dx - x e^{-x}\Big|_0^\infty = -e^{-x}\Big|_0^\infty = 1$$

1.13 Find volume of a sphere of Radius R

$$V = \int d\tau = \int_{r=0}^{R} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= \left(\int_0^R r^2 dr\right)\left(\int_0^\pi \sin\theta\, d\theta\right)\left(\int_0^{2\pi} d\phi\right)$$

$$= \left(\frac{R^3}{3}\right)(2)(2\pi) = \frac{4}{3}\pi R^3$$

1.14 Evaluate the integral $\int_0^3 x^3\, \delta(x-2)\, dx$

The delta function picks out the value of x^3 at point $x = 2$. so the integral is $2^3 = 8$

1.15 Show that $\delta(kx) = \frac{1}{|k|}\delta(x)$ where k is any (nonzero) constant

(In particular $\delta(-x) = \delta(x)$)

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For an arbitrary test function $f(x)$, consider the integral

$$\int_{-\infty}^{\infty} f(x)\,\delta(kx)\,dx.$$

$y = kx \rightarrow x = y/k \rightarrow dx = 1/k\,dy$

$$\int_{-\infty}^{\infty} f(x)\,\delta(kx)\,dx = \pm\int_{-\infty}^{\infty} f(y/k)\,\delta(y)\,\frac{dy}{k} = \pm\frac{1}{k}f(0) = \frac{1}{|k|}f(0)$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(kx)\,dx = \int_{-\infty}^{\infty} f(x)\left[\frac{1}{|k|}\delta(x)\right]dx$$

acc to eq. $\int_{-\infty}^{\infty} f(x)\,D_1(x)\,dx = \int_{-\infty}^{\infty} f(x)\,D_2(x)\,dx$] 2 expression involving delta function

so $\delta(kx)$ and $(1/|k|)\,\delta(x)$ are equal

1.16 Evaluate the integral $J = \int_V (r^2+2)\,\nabla\cdot\left(\frac{\hat{r}}{r^2}\right)d\tau$ where V is a sphere of radius R centered at origin

$$\nabla\cdot\left(\frac{\hat{r}}{r^2}\right) = 4\pi\delta^3(r)$$

$$\int_{\text{all space}} f(r)\,\delta^3(r-a)\,d\tau = f(a)$$

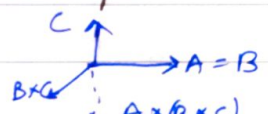
$$J = \int_V (r^2+2)\,4\pi\,\delta^3(r)\,d\tau = 4\pi\,(0+2) = 8\pi$$

Problems

1.2 Is the cross product associative? $(A\times B)\times C \overset{?}{=} A\times(B\times C)$ If so, prove it; if not, provide a counterexample

Is not in general perpendicular

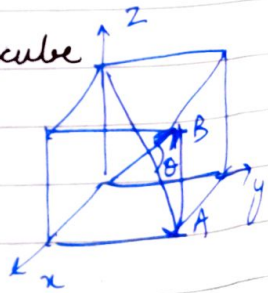
Eg. suppose $A = B$ and C is $\perp^r$ to A



$\therefore A\times(B\times C)$

Then $(B\times C)$ points out of page and $A\times(B\times C)$ points down and has magnitude ABC. But $(A\times B) = 0$ so $(A\times B)\times C = 0 \neq A\times(B\times C)$

1.3 Find the angle between the body diagonals of a cube



$A = +1\hat{x} + 1\hat{y} - 1\hat{z}$; $A = \sqrt{3}$;

$B = 1\hat{x} + 1\hat{y} + 1\hat{z}$; $B = \sqrt{3}$

$A\cdot B = +1+1-1 = 1 = AB\cos\theta = \sqrt{3}\sqrt{3}\cos\theta$

$\cos\theta = 1/3$

$\theta = \cos^{-1}(1/3) \approx 70.5^\circ$

1.5 Prove the BAC−CAB rule by writing out both sides in component form

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$$A \times (B \times C) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ (B_y C_z - B_z C_y) & (B_z C_x - B_x C_z) & (B_x C_y - B_y C_x) \end{vmatrix}$$

$$= \hat{x}\left|A_y (B_x C_y - B_y C_x) - A_z (B_z C_x - B_x C_z)\right| +$$
$$\hat{y}\left|-A_x (B_x C_y - B_y C_x) + A_z (B_y C_z - B_z C_y)\right| +$$
$$\hat{z}\left|A_x (B_z C_x - B_x C_z) - A_y (B_y C_z - B_z C_y)\right|$$

$$B(A \cdot C) - C(A \cdot B) = \left[B_x (A_x C_x + A_y C_y + A_z C_z) - C_x (A_x B_x + A_y B_y + A_z B_z)\right] \hat{x}$$
$$+ \left[B_y (A_x C_x + A_y C_y + A_z C_z) - C_y (A_x B_x + A_y B_y + A_z B_z)\right] \hat{y}$$
$$+ \left[B_z (A_x C_x + A_y C_y + A_z C_z) - C_z (A_x B_x + A_y B_y + A_z B_z)\right] \hat{z}$$

Checking the x component

$$A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z$$
$$= A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z$$

1.7 Find the separation vector $\mathbf{r}$ from the source point $(2, 8, 7)$ to the field point $(4, 6, 8)$. Determine its magnitude (r) and construct the unit vector $\hat{r}$.

$$\mathbf{r} = (4\hat{x} + 6\hat{y} + 8\hat{z}) - (2\hat{x} + 8\hat{y} + 7\hat{z}) = \boxed{2\hat{x} - 2\hat{y} + \hat{z}}$$

$$r = \sqrt{4 + 4 + 1} = 3$$

$$\hat{r} = \frac{\mathbf{r}}{r} = \frac{2}{3}\hat{x} - \frac{2}{3}\hat{y} + \frac{1}{3}\hat{z}$$

1.10 (a) How do the components of a vector transform under a translation of coordinates $(\bar{x} = x, \bar{y} = y - a, \bar{z} = z)$

No change $(\bar{A}_x = A_x, \bar{A}_y = A_y, \bar{A}_z = A_z)$

(b) How do the components of a vector transform under an inversion of coordinates $(\bar{x} = -x, \bar{y} = -y, \bar{z} = -z)$

$A \to -A \quad (\bar{A}_x = -A_x, \bar{A}_y = -A_y, \bar{A}_z = -A_z)$

(c) How do the component of a cross product transform under inversion? Is the cross product of 2 pseudovectors a vector or a pseudovector? Name 2 pseudovector quantities in classical mechanics

$(A \times B) \to (-A) \times (-B) = (A \times B)$

Cross product of 2 pseudovector is a pseudovector (no minus sign as compared to (b))

Angular momentum and torque

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(d) How does the scalar triple product of 3 vectors transform under inversion.

$A \cdot (B \times C) \rightarrow (-A) \cdot ((-B) \times (-C)) = -A \cdot (B \times C)$

So if $a = A \cdot (B \times C)$ then $\boxed{a \rightarrow -a}$ a pseudoscalar changes sign under inversion of coordinates.

1.11 Find the gradients of following function

(a) $f(x,y,z) = x^2 + y^3 + z^4$

$\nabla f = 2x\hat{x} + 3y^2\hat{y} + 4z^3\hat{z}$

(b) $f(x,y,z) = x^2 y^3 z^4$

$\nabla f = 2xy^3z^4\hat{x} + 3x^2y^3z^4\hat{y} + 4x^2y^3z^3\hat{z}$

(c) $f(x,y,z) = e^x \sin(y) \ln(z)$

$\nabla f = e^x \sin y \ln z\,\hat{x} + e^x \cos y \ln z\,\hat{y} + e^x \sin y \left(\frac{1}{z}\right)\hat{z}$

1.15 Calculate the divergence of following vector functions

(a) $V_a = x^2\hat{x} + 3xz^2\hat{y} - 2xz\hat{z}$

$\frac{\partial(x^2)}{\partial x} + \frac{\partial(3xz^2)}{\partial y} - \frac{\partial(-2xz)}{\partial z} = 2x + 0 - 2x = 0$

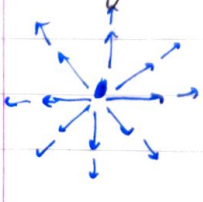
(b) $V_b = xy\hat{x} + 2yz\hat{y} + 3zx\hat{z}$

$\frac{\partial(xy)}{\partial x} + \frac{\partial(2yz)}{\partial y} + \frac{\partial(3xz)}{\partial z} = y + 2z + 3x$

(c) $V_c = y^2\hat{x} + (2xy + z^2)\hat{y} + 2yz\hat{z}$

$= \frac{\partial(y^2)}{\partial x} + \frac{\partial(2xy+z^2)}{\partial y} + \frac{\partial(2yz)}{\partial z} = 0 + 2x + 2y = 2(x+y)$

1.16 Sketch the vector function $V = \hat{r}/r^2$ and compute its divergence.



$V = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r^3}$

$\nabla \cdot V = \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right) = \frac{\partial}{\partial x}\left[x(x^2+y^2+z^2)^{-3/2}\right]$

$+ \frac{\partial}{\partial y}\left[y(x^2+y^2+z^2)^{-3/2}\right] + \frac{\partial}{\partial z}\left[z(x^2+y^2+z^2)^{-3/2}\right]$

$= (x^2+y^2+z^2)^{-3/2} + x(-3/2)(x^2+y^2+z^2)^{-5/2}(2x) + (x^2+y^2+z^2)^{-3/2}$

$+ y(-3/2)(x^2+y^2+z^2)^{-5/2}\,2y + (x^2+y^2+z^2)^{-3/2} +$

$z(-3/2)(x^2+y^2+z^2)^{-5/2}\,2z = 3r^{-3} - 3r^{-5}(x^2+y^2+z^2)$

$= 3r^{-3} - 3r^{-3} = 0$

$\nabla \cdot v$ is 0 everywhere except origin

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1.18 Calculate the curls of vector functions

a) $v_a = x^2\hat{x} + 3xz^2\hat{y} - 2xz\hat{z}$

$$\nabla \times v_a = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & 3xz^2 & -2xz \end{vmatrix} = \hat{x}(0-6xz) + \hat{y}(0+2z) + \hat{z}(3z^2-0)$$
$$= -6xz\hat{x} + 2z\hat{y} + 3z^2\hat{z}$$

b) $v_b = xy\hat{x} + 2yz\hat{y} + 3zx\hat{z}$

$$\nabla \times v_b = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & 2yz & 3zx \end{vmatrix} = \hat{x}(0-2y) + \hat{y}(0-3z) + \hat{z}(0-x)$$
$$= -2y\hat{x} - 3z\hat{y} - x\hat{z}$$

c) $v_c = y^2\hat{x} + (2xy + z^2)\hat{y} + 2yz\hat{z}$

$$\nabla \times v_c = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & (2xy+z^2) & 2yz \end{vmatrix} = \hat{x}(2z-2z) + \hat{y}(0-0) + \hat{z}(2y-2y)$$
$$= 0$$

$$\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$$

1.25 a) Check the product rule (by calculating each term separately) for

$A = x\hat{x} + 2y\hat{y} + 3z\hat{z}$ $\quad B = 3y\hat{x} - 2x\hat{y}$

$$A \times B = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & 2y & 3z \\ 3y & -2x & 0 \end{vmatrix} = \hat{x}(6xz) + \hat{y}(9zy) + \hat{z}(-2x^2 - 6y^2)$$

$$\nabla \cdot (A \times B) = \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z + 0 = 15z$$

$$\nabla \times A = \hat{x}\left(\frac{\partial}{\partial y}(3z) - \frac{\partial}{\partial z}(2y)\right) + \hat{y}\left(\frac{\partial}{\partial z}(x) - \frac{\partial}{\partial x}(3z)\right) + \hat{z}\left(\frac{\partial}{\partial z}(2y) - \frac{\partial}{\partial y}(x)\right)$$
$$= 0; \qquad B \cdot (\nabla \times A) = 0$$

$$\nabla \times B = \hat{x}\left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-2x)\right) + \hat{y}\left(\frac{\partial}{\partial z}(3y) - \frac{\partial}{\partial z}(0)\right) + \hat{z}\left(\frac{\partial}{\partial z}(-2x) - \frac{\partial}{\partial y}(3y)\right) = -5\hat{z}$$

$$A \cdot (\nabla \times B) = -15z$$

$$\nabla \cdot (A \times B) = B \cdot (\nabla \times A) - A \cdot (\nabla \times B) = 0 - (-15z) = 15z$$

b) Check $\nabla(A \cdot B) = A \times (\nabla \times B) + B \times (\nabla \times A) + (A \cdot \nabla)B + (B \cdot \nabla)A$

$A \cdot B = 3xy - 4xy = -xy$; $\quad \nabla(A \cdot B) = \nabla(-xy) = \hat{x}\frac{\partial}{\partial x}(-xy) + \hat{y}\frac{\partial}{\partial y}(-xy)$

$$= -y\hat{x} - x\hat{y}$$

$$A \times (\nabla \times B) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & 2y & 3z \\ 0 & 0 & -5 \end{vmatrix} = \hat{x}(-10y) + \hat{y}(5x); \quad B \times (\nabla \times A) = 0$$

$$(A \cdot \nabla)B = \left(x\frac{\partial}{\partial x} + 2y\frac{\partial}{\partial y} + 3z\frac{\partial}{\partial z}\right)(3y\hat{x} - 2x\hat{y}) = \hat{x}(6y) + \hat{y}(-2x)$$

$$(B \cdot \nabla)A = \left(3y\frac{\partial}{\partial x} - 2x\frac{\partial}{\partial y}\right)(x\hat{x} + 2y\hat{y} + 3z\hat{z}) = \hat{x}(3y) + \hat{y}(-4x)$$

$$A \times (\nabla \times B) + B \times (\nabla \times A) + (A \cdot \nabla)B + (B \cdot \nabla)A$$
$$= -10y\hat{x} + 5x\hat{y} + 6y\hat{x} - 2x\hat{y} + 3y\hat{x} - 4x\hat{y}$$
$$= -y\hat{x} - x\hat{y} = \nabla \cdot (A \cdot B)$$

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c) Check $\nabla\times(A\times B) = (B\cdot\nabla)A - (A\cdot\nabla)B + A(\nabla\cdot B) - B(\nabla\cdot A)$

$\nabla\times(A\times B) = \hat{x}\left(\frac{\partial}{\partial y}(-2x^2-6y^2) - \frac{\partial}{\partial z}(9zy)\right) + \hat{y}\left(\frac{\partial}{\partial z}(6xz) - \frac{\partial}{\partial x}(-2x^2-6y^2)\right) + \hat{z}\left(\frac{\partial}{\partial x}(9zy) - \frac{\partial}{\partial y}(6xz)\right)$

$= \hat{x}(-12y - 9y) + \hat{y}(6x + 4x) + \hat{z}(0) = \underline{-21y\,\hat{x} + 10x\,\hat{y}}$

$\nabla\cdot A = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(3z) = 1+2+3 = 6$

$\nabla\cdot B = \frac{\partial}{\partial x}(3y) + \frac{\partial}{\partial y}(-2x) = 0$

$(B\cdot\nabla)A - (A\cdot\nabla)B + A(\nabla\cdot B) - B(\nabla\cdot A) = 3y\,\hat{x} - 4x\,\hat{y} - 6y\,\hat{x} + 2x\,\hat{y} - 18y\,\hat{x} + 12x\,\hat{y} = \underline{-21y\,\hat{x} + 10x\,\hat{y}} = \nabla\times(A\times B)$

1.26 Calculate the Laplacian of the following

(a) $T_a = x^2 + 2xy + 3z + 4$

$\frac{\partial^2 T_a}{\partial x^2} = 2, \quad \frac{\partial^2 T_a}{\partial y^2} = \frac{\partial^2 T_a}{\partial z^2} = 0 \Rightarrow \boxed{\nabla^2 T_a = 2}$

(b) $T_b = \sin x \sin y \sin z$

$\frac{\partial^2 T_b}{\partial x^2} = \frac{\partial^2 T_b}{\partial y^2} = \frac{\partial^2 T_b}{\partial z^2} = -T_b \Rightarrow \boxed{\nabla^2 T_b = -3T_b = -3\sin x \sin y \sin z}$

(c) $T_c = e^{-5x}\sin 4y \cos 3z$

$\frac{\partial^2 T_c}{\partial x^2} = 25\,T_c, \quad \frac{\partial^2 T_c}{\partial y^2} = -16\,T_c, \quad \frac{\partial^2 T_c}{\partial z^2} = -9\,T_c \Rightarrow \boxed{\nabla^2 T_c = 0}$

(d) $v = x^2\hat{x} + 3xz^2\hat{y} - 2xz\hat{z}$

$\frac{\partial^2 v_x}{\partial x^2} = 2, \quad \frac{\partial^2 v_x}{\partial y^2} = \frac{\partial^2 v_x}{\partial z^2} = 0 \Rightarrow \nabla^2 v_x = 2$

$\frac{\partial^2 v_y}{\partial x^2} = \frac{\partial^2 v_y}{\partial y^2} = 0, \quad \frac{\partial^2 v_y}{\partial z^2} = 6x \Rightarrow \nabla^2 v_y = 6x$

$\frac{\partial^2 v_z}{\partial x^2} = \frac{\partial^2 v_z}{\partial y^2} = \frac{\partial^2 v_z}{\partial z^2} = 0 \Rightarrow \nabla^2 v_z = 0$

$\boxed{\nabla^2 v = 2\hat{x} + 6x\,\hat{y}}$

1.32 Check the fundamental theorem for gradients using $T = x^2 + 4xy + 2yz^3$ the points $a = (0,0,0)$, $b = (1,1,1)$ and the three paths

(a) $(0,0,0) \longrightarrow (1,0,0) \longrightarrow (1,1,0) \longrightarrow (1,1,1)$

$T(b) = 1+4+2 = 7, \; T(a) = 0 \Rightarrow \boxed{T(b) - T(a) = 7}$

$\nabla T = (2x+4y)\hat{x} + (4x+2z^3)\hat{y} + (6yz^2)\hat{z}$

$\nabla T\,dl = (2x+4y)\,dx + (4x+2z^3)\,dy + (6yz^2)\,dz$

Segment 1: $x: 0\to 1, \; y = z = dy = dz = 0. \quad \int \nabla T\cdot dl = \int_0^1 (2x)\,dx = x^2\big|_0^1 = 1$

Segment 2: $y: 0\to 1, \; x = 1, \; z = 0, \; dx = dz = 0. \quad \int \nabla T\cdot dl = \int_0^1 4\,dy = 4y\big|_0^1 = 4$

Segment 3: $z: 0\to 1, \; x = y = 1, \; dx = dy = 0 \quad \int \nabla T\cdot dl = \int_0^1 6z^2\,dz = 2z^3\big|_0^1 = 2$

$\int_a^b \nabla T\cdot dl = 7$

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(b) $(0,0,0) \to (0,0,1) \to (0,1,1) \to (1,1,1)$

Segment 1: $z: 0 \to 1$, $x = y = dx = dy = 0$ $\int \nabla T \cdot dl = \int_0^1 (0)\, dz = 0$

Segment 2: $y: 0 \to 1$, $x = 0$, $z = 1$, $dx = dz = 0$ $\int \nabla T \cdot dl = \int_0^1 2\, dy = 2y\big|_0^1 = 2$

Segment 3: $x: 0 \to 1$, $y = z = 1$, $dy = dz = 0$ $\int \nabla \cdot T \cdot dl = \int_0^1 (2x+4)\, dx = (x^2 + 4x)\big|_0^1$

$= 5$

(c) the parabolic path $z = x^2$; $y = x$ $\quad x: 0 \to 1$, $y = x$, $z = x^2$, $dy = dx$, $dz = 2x\, dx$

$\nabla T \cdot dl = (2x + 4x)\, dx + (4x + 2x^6)\, dx + (6x x^4)\, 2x\, dx$

$= (10x + 14x^6)\, dx$

$\int_a^b \nabla T \cdot dl = \int_0^1 (10x + 14x^6)\, dx = (5x^2 + 2x^7)\big|_0^1 = 5 + 2 = 7$

1.33 Test the divergence theorem for the function $v = (xy)\hat{x} + (2yz)\hat{y} + (3zx)\hat{z}$, Take as your volume the cube with side length 2

$\nabla \cdot v = y + 2z + 3x$

$\int (\nabla \cdot v)\, d\tau = \int (y + 2z + 3x)\, dx\, dy\, dz = \int_0^2 \int_0^2 \int_0^2 (y + 2z + 3x)\, dx\, dy\, dz$

$= \int \{ \int_0^2 (2y + 4z + 6)\, dy \}\, dz$

$= \int_0^2 (8z + 16)\, dz = (4z^2 + 16z)\big|_0^2 = 16 + 32 = 48$

(i) $da = dy\, dz\, \hat{x}$, $x = 2$, $v \cdot da = 2y\, dy\, dz$, $\int v \cdot da = \int\int 2y\, dy\, dz = 2y^2\big|_0^2 = 8$

(ii) $da = -dy\, dz\, \hat{x}$, $x = 0$, $v \cdot da = 0$, $\int v \cdot da = 0$

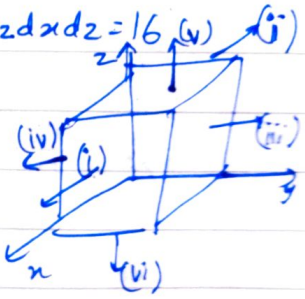
(iii) $da = dx\, dz\, \hat{y}$, $y = 2$, $v \cdot da = 4z\, dx\, dz$, $\int v \cdot da = \int\int 4z\, dx\, dz = 16$

(iv) $da = -dx\, dz\, \hat{y}$, $y = 0$, $v \cdot da = 0$, $\int v \cdot da = 0$

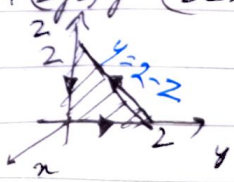
(v) $da = dx\, dy\, \hat{z}$, $z = 2$, $v \cdot da = 6x\, dx\, dy$, $\int v \cdot da = 24$

(vi) $da = -dx\, dy\, \hat{z}$, $z = 0$, $v \cdot da = 0$, $\int v \cdot da = 0$

$\Rightarrow \int v \cdot da = 8 + 16 + 24 = 48$



1.34 Test Stokes' theorem for function $v = (xy)\hat{x} + (2yz)\hat{y} + (3zx)\hat{z}$ using triangular shaded area of Fig.



$\nabla \times v = \hat{x}(0 - 2y) + \hat{y}(0 - 3z) + \hat{z}(0 - x)$

$= -2y\hat{x} - 3z\hat{y} - x\hat{z}$

$da = dy\, dz\, \hat{x}$

$(\nabla \times v) \cdot da = -2y\, dy\, dz$

$\int (\nabla \times v) \cdot da = \int \{ \int_0^{2-z} (-2y)\, dy \}\, dz$

$= -\int_0^2 (4 - 4z + z^2)\, dz = -(4z - 2z^2 + \frac{z^3}{3})\big|_0^2$

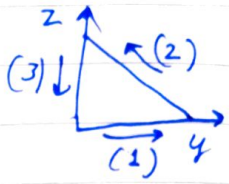
$= -(8 - 8 + \frac{8}{3}) = -\frac{8}{3}$

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$v \cdot dl = (xy)\,dx + (2yz)\,dy + (3zx)\,dz$

There are 3 segment



(1) $x = z = 0$; $dx = dz = 0$, $y : 0 \to 2$, $\int v \cdot dl = 0$

(2) $x = 0$; $z = 2 - y$; $dx = 0$, $dz = -dy$,

$y : 2 \to 0$, $v \cdot dl = 2yz\,dy$

$$\int v \cdot dl = \int_2^0 2y(2-y)\,dy = -\int_0^2 (4y - 2y^2)\,dy$$

$$= -(2y^2 - \tfrac{2}{3}y^3)\Big|_0^2 = -(8 - \tfrac{2}{3} \times 8) = -8/3$$

(3) $x = y = 0$; $dx = dy = 0$; $z : 2 \to 0$, $v \cdot dl = 0$.

$\int v \cdot dl = 0$

$\oint v \cdot dl = -8/3$

1.44 Evaluate the following integrals

(a) $\int_2^6 (3x^2 - 2x - 1)\,\delta(x-3)\,dx$

$3(3^2) - 2(3) - 1 = 27 - 6 - 1 = 20$

(b) $\int_0^5 \cos x\,\delta(x-\pi)\,dx$

$\cos \pi = -1$

(c) $\int_0^3 x^3\,\delta(x+1)\,dx = 0$ (outside range)

(d) $\int_{-\infty}^{\infty} \ln(x+3)\,\delta(x+2)\,dx$

$\ln(-2+3) - \ln 1 = 0$

1.49 Evaluate the integral $J = \int_{\mathcal{V}} e^{-r} \left(\nabla \cdot \frac{\hat{r}}{r^2}\right) d\tau$ (where $\mathcal{V}$ is a sphere of radius R centered at origin) by 2 diff methods

First method.

$J = \int e^{-r} (4\pi \delta^3(r))\,d\tau = 4\pi e^{-0} = 4\pi$

Second method

$$J = -\int_{\mathcal{V}} \frac{\hat{r}}{r^2} \nabla(e^{-r})\,d\tau + \oint_S e^{-r} \frac{\hat{r}}{r^2} \cdot da$$

$$\nabla(e^{-r}) = \left(\frac{\partial}{\partial r} e^{-r}\right)\hat{r} = -e^{-r}\hat{r}$$

$$J = \int \frac{1}{r^2} e^{-r} 4\pi r^2\,dr + \int e^{-r} \frac{\hat{r}}{r^2} r^2 \sin\theta\,d\theta\,d\phi\,\hat{r}$$

$$= 4\pi \int_0^R e^{-r}\,dr + e^{-R} \int \sin\theta\,d\theta\,d\phi$$

$$= 4\pi(-e^{-r})\Big|_0^R + 4\pi e^{-R}$$

$$= 4\pi(-e^{-R} + e^{-0}) + 4\pi e^{-R} = 4\pi$$

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1.50 a) Let $F_1 = x^2 \hat{z}$ and $F_2 = x\hat{x} + y\hat{y} + z\hat{z}$. Calculate the divergence and curl of F_1 and F_2. Which one can be written as the gradient of a scalar? Find a scalar potential that does the job. Which one can be written as the curl of a vector? Find a suitable vector potential.

$$\nabla \cdot F_1 = \frac{\partial(0)}{\partial x} + \frac{\partial(0)}{\partial y} + \frac{\partial(x^2)}{\partial z} = 0 \quad ; \quad \nabla \cdot F_2 = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1+1+1 = 3$$

$$\nabla \times F_1 = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & 0 & x^2 \end{vmatrix} = -\hat{y}\frac{\partial}{\partial x}(x^2) = -2x\hat{y}$$

$$\nabla \times F_2 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = 0$$

F_2 is a gradient, F_1 is a curl

$U_2 = \frac{1}{2}(x^3 + y^2 + z^2)$ would do $(F_2 = \nabla U_2)$

For A_1, we want $\left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y}\right) = \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) = 0$;

$\partial A_y/\partial x - \partial A_y/\partial y = x^2 \quad A_y = x^3/3, \; A_x = A_z = 0$

$A_1 = 1/3\, x^2 \hat{y} \quad (F_1 = \nabla \times A_1)$

b) Show that $F_3 = yz\hat{x} + zx\hat{y} + xy\hat{z}$ can be written both as the gradient of a scalar and as the curl of a vector. Find scalar and vector potentials for this function.

$$\nabla \cdot F_3 = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy) = 0$$

$$\nabla \times F_3 \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz & xy \end{vmatrix} = \hat{x}(x-x) + \hat{y}(y-y) + \hat{z}(z-z) = 0$$

$F_3 = \nabla U_3 = \nabla \times A_3$

$U_3 = xyz$

$\partial A_z/\partial y - \partial A_y/\partial z = yz, \quad A_z = \frac{1}{4}y^2 z + f(x,z); \; A_y = -\frac{1}{4}y^2 z + g(x,y)$

$\partial A_x/\partial z - \partial A_z/\partial x = xz, \quad A_x = \frac{1}{4}z^2 y + h(x,y); \; A_z = -\frac{1}{4}zx^2 + j(y,z)$

$\partial A_y/\partial x - \partial A_x/\partial y = xy, \quad A_y = \frac{1}{4}x^2 y + k(y,z); \; A_x = -\frac{1}{4}xy^2 + l(x,z)$

$A_3 = \frac{1}{4}\{x(z^2 - y^2)\hat{x} + y(x^2 - z^2)\hat{y} + z(y^2 - x^2)\hat{z}\}$

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Chapter 2: Electrostatics

Q1. What is Coulomb's ~~expee~~ law ? Write its expression.

The force on a test charge Q due to a single point charge q, that is at rest a distance r away is proportional to the product of charges and inversely proportional to the square of the separation of distance

$$F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}$$

Q2. What is an electric field ? Write its expression

The electric field at a point is defined as the electric force experienced by a unit positive test charge placed at that point

$$\vec{E} = \vec{F}/q$$

$$E(r) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i^2} \hat{r}$$

Q3. What is Gauss's law ? Write its expression

Gauss's law states that the total electric flux through any closed surface is equal to the total charge enclosed by the surface ~~divided by~~ ϵ_0

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

Integral form

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Differential form

Q4. What is electric potential ? Write its expression

Electric potential at a point is the work done per unit +ve test charge in bringing the charge from infinity to that point

$$V = W/q \qquad \vec{E} = -\nabla V$$

$$V(r) = -\int_{\infty}^{r} E \cdot dl$$

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Q5. What are Poisson's equation and Laplace's equation?

These equations describe the relationship between electric potential and charge distribution

Poisson's Equation: $\boxed{\nabla^2 V = \frac{-\rho}{\epsilon_0}}$ applies in a region where charge density ρ is present

Laplace's Equation: $\boxed{\nabla^2 V = 0}$ In a charge-free region

Q6. Explain boundary conditions in Electrostatics.

Boundary conditions tells us how the electric field and potential behave at the boundary between 2 regions

$$V_1 = V_2$$

$$E^{\parallel}_{above} = E^{\parallel}_{below}$$

$$E^{\perp}_{above} - E^{\perp}_{below} = \sigma/\epsilon_0$$

Q7. Describe the expression of the work it takes to move a charge

To move a charge Q from a to b

$$W = \int_a^b F \cdot dl = -Q\int_a^b E \cdot dl = Q\,[V(b) - V(a)]$$

$$W = QV(r)$$

Q8. Describe the expression of Energy for a point charge distribution

The first charge q_1 takes no work, since there is no field yet to fight

For second charge $W = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}}$

For many pt. charges $U = \frac{1}{4\pi\epsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}}$

$$U = \frac{1}{2}\sum_i q_i V_i$$

Q9. Describe the expression of Energy for a continuous charge distribution

For a volume charge density ρ

$$W = \tfrac{1}{2}\int \rho V \, d\tau$$

$$W = \frac{\epsilon_0}{2}\int E^2 \, d\tau$$

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Q10. Describe the properties of conductors

(i) $E=0$ inside a conductor
(ii) $\rho=0$ inside a conductor
(iii) net charge resides on a surface
(iv) A conductor is an equipotential
(v) E is perpendicular to the surface, just outside a conductor

Q11. Explain the properties of capacitors.

A capacitor is a device used to store electric charge and electrostatic energy. It consists of 2 conductors separated by an insulating material.

$C = Q/V$

Energy stored in a capacitor: $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$

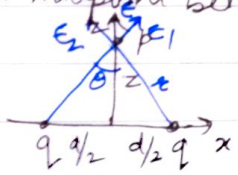
Parallel Plate capacitor $C = A\epsilon_0/d$

Capacitance in parallel $C_{eq} = C_1 + C_2 + C_3 + \cdots$

Capacitance in series $1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \cdots$

Examples

2.1 Find the electric field at distance z above the midpoint between 2 equal charges (q), a distance d apart



Let E_1 be the field of left charge alone,

E_2 that of right charge alone

Horizontal component cancels out

$$E_2 = 2 \times \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\cos\theta$$

$$r = \sqrt{z^2 + (d/2)^2} \qquad \cos\theta = z/r$$

$$E = \frac{1}{4\pi\epsilon_0}\frac{2qz}{[z^2 + (d/2)^2]^{3/2}}\hat{z}$$

2.3 Find the field outside a uniformly charged solid sphere of radius R and total charge q

Imagine a spherical surface $r > R$

$$\oint_S E \cdot da = \frac{Q_{enc}}{\epsilon_0}$$

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$$\int_S E \cdot da = \int_S |E|\, da$$
$$\int_S |E|\, da = |E| \int_S da = |E|\, 4\pi r^2$$
$$|E|\, 4\pi r^2 = q/\epsilon_0$$
$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$$

2.5 An infinite plane carries a uniform surface charge σ. Find its electric field.

Gaussian pillbox extending equal distance above & below plane

$$\oint E \cdot da = Q_{enc}/\epsilon_0$$

$Q_{enc} = \sigma A$ A = area of lid of pillbox



By symmetry $\int E \cdot da = 2A|E|$

$$2A|E| = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}\hat{n}$$

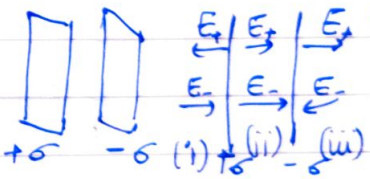
2.6 Two infinite parallel planes carry equal but opposite uniform charge densities $\pm\sigma$. Find the field in each of the 3 regions (i) left of both (ii) between them (iii) right of both

Left plate produces a field $(1/2\epsilon_0)\sigma$ which points away from it. to the left in region (i) and right in region (ii) & (iii)

The right plate produces $(1/2\epsilon_0)\sigma$ toward it, to right in region (i) and (ii) and left in region (iii)

Field between plate is σ/ϵ_0

0 to the left and right of both



2.7 Find the potential inside and outside a spherical shell of radius R that carries a uniform surface charge. Set reference point at infinity

$\frac{q}{r}$

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Field outside is $E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$

where q is total charge on sphere. Field inside is 0.

For points outside the sphere $V(r) = -\int_0^r E\,dl = \frac{-1}{4\pi\epsilon_0}\int_\infty^r \frac{q}{r'^2}dr' = \frac{1}{4\pi\epsilon_0}\frac{q}{r'}\Big|_\infty^r = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$

Potential inside sphere

$$V(r) = \frac{-1}{4\pi\epsilon_0}\int_\infty^R \frac{q}{r'^2}dr' - \int_R^r (0)\,dr' = \frac{1}{4\pi\epsilon_0}\frac{q}{r'}\Big|_\infty^R + 0 = \frac{1}{4\pi\epsilon_0}\frac{q}{R}$$

2.9 Find the energy of a uniformly charged spherical shell of total charge q and radius R

Sol^n 1. $W = \frac{1}{2}\int \sigma V\,da \qquad V = \frac{1}{4\pi\epsilon_0}\frac{q}{R}$

$$W = \frac{1}{8\pi\epsilon_0}\frac{q}{R}\int \sigma\,da = \frac{1}{8\pi\epsilon_0}\frac{q^2}{R}$$

Sol^n 2. Inside sphere $E = 0$

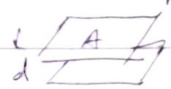
$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r} \quad \text{so} \quad E^2 = \frac{q^2}{(4\pi\epsilon_0)^2 r^4}$$

$$W_{tot} = \frac{\epsilon_0}{2(4\pi\epsilon_0)^2}\int_{outside}\left(\frac{q^2}{r^4}\right)(r^2\sin\theta\,dr\,d\theta\,d\phi)$$

$$= \frac{1}{32\pi^2\epsilon_0}q^2 4\pi\int_R^\infty \frac{1}{r^2}dr = \frac{1}{8\pi\epsilon_0}\frac{q^2}{R}$$

2.11 Find the capacitance of a parallel plate capacitor consisting of 2 metal surfaces of area A held at a distance d apart

If we put $+Q$ on top & $-Q$ on bottom, they will spread out uniformly. Surface charge density is ~~$\sigma = Q/A$~~ $\sigma = Q/A$ on top of plat



So field $= \sigma/\epsilon_0 = (1/\epsilon_0)(Q/A)$

Potential diff $V = \frac{Q\,d}{A\epsilon_0}$

$$C = Q/V = \frac{A\epsilon_0}{d}$$

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2.12 Find the capacitance of 2 concentric spherical metal shells with radii a and b

Place charge $+Q$ on inner sphere and $-Q$ on outer one

Field between spheres is $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}$

Potential diff

$$V = -\int_b^a E \cdot dl = -\frac{Q}{4\pi\epsilon_0}\int_b^a \frac{1}{r^2}dr = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$C = \frac{Q}{V}$$

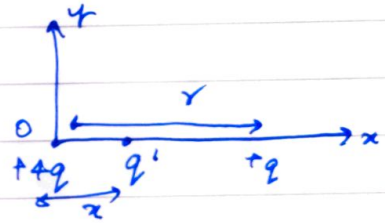
$$= 4\pi\epsilon_0 \frac{ab}{b-a}$$

Handwritten

Q24 Two free charges $+4q$ and $+q$ are placed at a distance r apart. find the coulomb force btw them. Also find magnitude, sign & location of 3rd charge q' such that

(i) no force on q'

(ii) No force on either of the charge $+4q$, $+q$ & q'. This is called Entire system is in equilibrium



$$\vec{F}_{+4q} = \frac{1}{4\pi\epsilon_0}\frac{4q\cdot q}{r^2}\hat{r} = \frac{1}{4\pi\epsilon_0}\frac{4q^2}{r^2}(-\hat{x})$$

$$\vec{F}_q = \frac{1}{4\pi\epsilon_0}\frac{4q^2}{r^2}(+\hat{x})$$

(i) Suppose $q' = +ve$

$$\vec{F}_{q'} = \frac{1}{4\pi\epsilon_0}\frac{4q\cdot q'}{x^2}(+\hat{x}) \quad —(1)$$

$$\vec{F}_{q'} = \frac{1}{4\pi\epsilon_0}\frac{qq'}{(r-x)^2}(-\hat{x}) \quad —(2)$$

$$\vec{F}_{q'} + \vec{F}_{q'} = 0$$

$$\frac{1}{4\pi\epsilon_0}\frac{4qq'}{x^2}\hat{x} = \frac{1}{4\pi\epsilon_0}\frac{qq'}{(r-x)^2}(-\hat{x})$$

$$\Rightarrow 4(r-x)^2 = x^2$$

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$$4r^2 + 4x^2 - 8rx = x^2$$

$$3x^2 - 8rx + 4r^2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{8r \pm \sqrt{64r^2 - 48r^2}}{6} = \frac{8r + 4r}{6}$$

$$= 2r \text{ or } 2r/3$$

$x = 2r$ not acceptable so $x = 2r/3$

Now if q' is -ve then forces b/w $4q$ & q' and also b/w q' & q is attractive. Same answer in this case as well

(ii) If q is +ve equilibrium cannot be acheived in entire system ~~as~~.

If q = -ve then zero force cond^n can be acheived

$$\vec{F}_{4q} = \frac{1}{4\pi\epsilon_0} \frac{4qq'}{x^2} (+\hat{x})$$

$$\vec{F}_{4q} = \frac{1}{4\pi\epsilon_0} \frac{4qq}{r^2} (-\hat{x})$$

$$\vec{F}_{4q} + \vec{F}_{4q} = 0$$

$$\frac{1}{4\pi\epsilon_0} \frac{4qq'}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{4qq}{r^2}$$

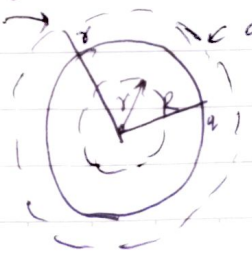
$$q'/x^2 = q/r^2$$

$$q' = \frac{q}{r^2} \times x^2 = \frac{q}{r^2} \times \frac{4r^2}{q}$$

$$q' = \frac{-4q}{q}$$

If we put q' at $4q/q$ distance then there will be no force on entire system

Q6) Find the potential inside & outside a conducting sphere of radius R & charge q.



Pot^n $V_{out} = -\int \vec{E} \cdot d\vec{r}$

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Outside Electric field $\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$

Charge is not extended to ∞ itself so we'll take ∞ as reference

$$V_{out} = -\int_{\infty}^{r} E \cdot dr$$

$$= -\int_{\infty}^{r} \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{-q}{4\pi\epsilon_0}\left[\frac{-1}{r}\right]_{\infty}^{r}$$

$$\boxed{V_{out} = \frac{q}{4\pi\epsilon_0 r}}$$

V_{in} → Work done to take a unit test charge from ∞ to surface & surface to r

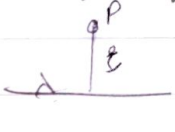
$$V_{in} = -\int_{\infty}^{r} E\, d\vec{r}$$

$$= -\int_{\infty}^{R} \frac{q}{4\pi\epsilon_0 r^2} dr - \int_{R}^{r} 0\, dr$$

$$\boxed{V_{in} = \frac{q}{4\pi\epsilon_0 R}}$$

63 Find the potential at a distance s from an infinitely long straight wire that carries uniform line charge λ

at P, $\vec{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{s}$ $\quad V = -\int_{a}^{s} E \cdot ds$



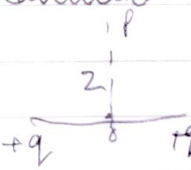
$$V = -\int_{a}^{s} \frac{\lambda}{2\pi\epsilon_0 s} ds$$

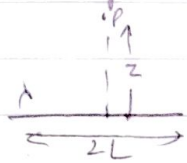
$$= \frac{-\lambda}{2\pi\epsilon_0}(\ln s)_a^s = \frac{-\lambda}{2\pi\epsilon_0}\left(\ln \frac{s}{a}\right)$$

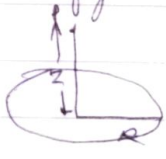
$$V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{s}\right)$$

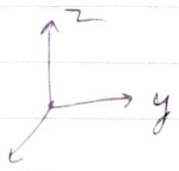
a → finite pt. at which we know potential reference point

65 Calculate the potential of 3 charge configurations given below





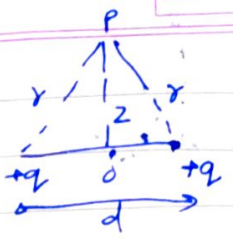




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(i)



$$V_2 = V_1 = q/4\pi\epsilon_0 r$$

$$= \frac{q}{4\pi\epsilon_0\sqrt{z^2+d^2/4}}$$

$$V = V_1 + V_2$$

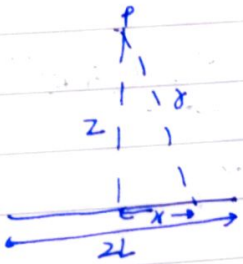
$$V = \frac{2q}{4\pi\epsilon_0\sqrt{z^2+d^2/4}} \Rightarrow \boxed{V = \frac{q}{2\pi\epsilon_0\sqrt{z^2+d^2/4}}}$$

Elec field $\vec{E} = -\vec{\nabla}V = -\partial V/\partial z\ \hat{z}$

$$= \frac{-q}{2\pi\epsilon_0} \times \frac{1}{2}\left(z^2+d^2/4\right)^{-3/2}(2z)$$

$$\vec{E} = \frac{qz}{2\pi\epsilon_0\left(z^2+\frac{d^2}{4}\right)^{3/2}}\hat{z}$$

(ii)



$$r = \sqrt{z^2+x^2} \qquad V = \frac{1}{4\pi\epsilon_0}\int\frac{\lambda dl}{r}$$

$$V = \frac{2}{4\pi\epsilon_0}\int_0^L \frac{\lambda dx}{\sqrt{z^2+x^2}}$$

$$V = \frac{\lambda}{2\pi\epsilon_0}\ln\left[x+\sqrt{z^2+x^2}\right]_0^L$$

{ Using $\int\frac{dx}{(a^2+x^2)^{1/2}} = \ln\left[x+\sqrt{a^2+x^2}\right]$ }

$$V = \frac{\lambda}{2\pi\epsilon_0}\ln\left[\frac{L+\sqrt{L^2+z^2}}{z}\right]$$

$$\vec{E} = -\partial V/\partial z\ \hat{z}$$

$$= \frac{-\partial}{\partial z}\left[\frac{\lambda}{2\pi\epsilon_0}\left(\ln\left\{L+\sqrt{z^2+L^2}\right\} - \ln z\right)\right]$$

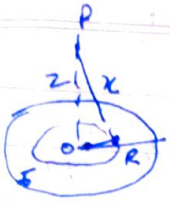
$$= \frac{-\lambda}{2\pi\epsilon_0}\left[\frac{-2z}{(L+\sqrt{L^2+z^2})2\cdot\sqrt{z^2+L^2}} - \frac{1}{z}\right]\hat{z}$$

$$= \frac{-\lambda}{2\pi\epsilon_0}\left[\frac{z}{(L+\sqrt{z^2+L^2})\sqrt{z^2+L^2}} - \frac{1}{z}\right]\hat{z}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0}\left[\frac{1}{z} - \frac{z}{(L+\sqrt{z^2+L^2}+z^2+L^2)}\right]\hat{z}$$

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(iii)



$dq = 2\pi r\,dr\,\sigma$

$$dV = \frac{1}{4\pi\epsilon_0}\frac{dq}{\sqrt{z^2+r^2}}$$

$$x = \sqrt{z^2+r^2}$$

$$V = \frac{\sigma}{4\pi\epsilon_0}\int_0^R \frac{2\pi r\,dr}{(z^2+r^2)^{1/2}}$$

$$z^2 + r^2 = t$$

$$2r\,dr = dt$$

$$V = \frac{\sigma}{4\pi\epsilon_0}\,2\pi\int\frac{dt}{2(t)^{1/2}}$$

$$V = \frac{\sigma}{4\epsilon_0}\left[\frac{t^{1/2}}{1/2}\right]$$

$$= \frac{\sigma}{2\epsilon_0}\left[(z^2+r^2)^{1/2}\right]_0^R \Rightarrow \boxed{V = \frac{\sigma}{2\epsilon_0}\left[(z^2+R^2)^{1/2} - z\right]}$$

$$\boxed{V_{centre} = \frac{\sigma R}{2\epsilon_0}} \quad \boxed{\text{at centre} \rightarrow z = 0}$$

$$V_{edge} = \sigma R/\pi\epsilon_0$$

Q6 Find whether the electric field $\vec{E} = (12x^2 - y^2)\hat{x} - 2xy\hat{y}$ is conservative or not. If conservative find the potential as a func^n of position

$\vec{E} = (12x^2 - y^2)\hat{x} - 2xy\hat{y}$

$\vec{E}$ will be conservative if $\vec{\nabla}\times\vec{E} = 0$

$$\vec{\nabla}\times\vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (12x^2 - y^2) & -2xy & 0 \end{vmatrix}$$

$$= \hat{x}[0] - \hat{y}[0] + \hat{z}[-2y + 2y]$$

$\vec{\nabla}\times E = 0$ so $\vec{E}$ is conservative

$$V = -\int E\cdot dr$$

$$= -\int d(4x^3 - xy^2)$$

$$V = xy^2 - 4x^3 + C$$

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Q98 The plates of a parallel plate capacitor are located at $x=0$ & $x=L$. The plate at $x=0$ is grounded while plate at $x=L$ is at potential V. The space between the plate has a uniform charge density ρ. Find potential ~~but~~ & Elec field b/w plates

By poisson's equation

$$\nabla^2 V = -\rho/\epsilon_0$$

$$d^2V/dx^2 = -\rho/\epsilon_0$$

$$dV/dx = -\rho x/\epsilon_0 + A$$

$$V = -\rho/\epsilon_0 \frac{x^2}{2} + Ax + B$$

$x=0,\ V=0 \Rightarrow 0 = 0+0+B \Rightarrow B=0$

$x=L,\ V=V_0 \Rightarrow V_0 = -\frac{\rho}{\epsilon_0}\frac{L^2}{2} + AL + 0 \Rightarrow AL = V_0 + \frac{\rho_0 L^2}{2\epsilon_0}$

$$\boxed{A = \frac{V_0}{L} + \frac{\rho_0 L}{2\epsilon_0}}$$

So $$V = -\frac{\rho x^2}{2\epsilon_0} + \left(\frac{V_0}{L} + \frac{\rho_0 L}{2\epsilon_0}\right)x$$

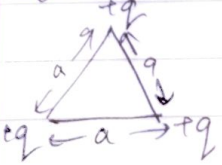
or $$\boxed{V = -\frac{\rho x^2}{2\epsilon_0} + \frac{V_0 x}{L} + \frac{\rho_0 L x}{2\epsilon_0}}$$

$$\vec{E} = -\vec{\nabla} V = \partial V/\partial x\ \hat{x}$$

$$= -\frac{\partial}{\partial x}\left[-\frac{\rho x^2}{2\epsilon_0} + \frac{V_0 x}{L} + \frac{\rho_0 L x}{2\epsilon_0}\right]\hat{x}$$

$$\vec{E} = \left[\frac{\rho x}{\epsilon_0} - \frac{V_0}{L} - \frac{\rho L}{2\epsilon_0}\right]\hat{x}$$

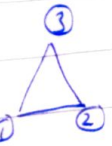
Q81 Calculate energy of this config.



Energy of system of 3 charges will be

$$W = \frac{1}{8\pi\epsilon_0}\sum_{i=1}^{3}\sum_{j=1}^{3}\frac{q_i q_i}{r_{ij}}$$

$$W = \frac{1}{8\pi\epsilon_0}\left[\frac{q_1 q_2}{a} + \frac{q_2 q_3}{a} + \frac{q_3 q_1}{a}\right]$$



$$W = \frac{1}{8\pi\epsilon_0}\left[\frac{6q^2}{a}\right]$$

$$W = \frac{3q^2}{4\pi\epsilon_0 a}$$

$W_1 = 0$

$W_2 = qV_1 = q^2/4\pi\epsilon_0 a$

$W_3 = qV_1 + qV_2 = \frac{q^2}{4\pi\epsilon_0 a} + \frac{q^2}{4\pi\epsilon_0 a}$

$W = W_1 + W_2 + W_3$

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Chapter 3 : Potentials

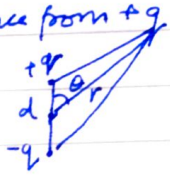
Theory

Q1. Explain the concept of the method of images in finding electric potential.

A charge $+q$ placed near a grounded conducting plane creates an induced charge distribution on the conductor. Instead of calculating these induced charges directly, we replace their effect by an imaginary image charge $-q$ placed symmetrically behind the conducting plane. The image charge is not real charge. It is a mathematical calculation used to satisfy the boundary condition on the grounded conductor.

Q2. ~~Explain the concept~~ Using multipole expansion, find ~~approx~~ approx potential at pts far from dipole

Let r_- be the distance from $-q$ & r_+ be distance from $+q$



$$V(r) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_+} - \frac{q}{r_-}\right)$$

and $$r_\pm^2 = r^2 + (d/2)^2 \mp rd\cos\theta$$

$$= r^2\left(1 \mp \frac{d}{r}\cos\theta + \frac{d^2}{4r^2}\right)$$

$r \gg d$

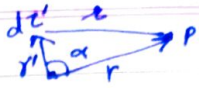
$$\frac{1}{r_\pm} \simeq \frac{1}{r}\left(1 \mp \frac{d}{r}\cos\theta\right)^{-1/2} \simeq \frac{1}{r}\left(1 \pm \frac{d}{2r}\cos\theta\right)$$

$$\frac{1}{r_+} - \frac{1}{r_-} \simeq \frac{d}{r^2}\cos\theta$$

$$V(r) \simeq \frac{1}{4\pi\epsilon_0}\frac{qd\cos\theta}{r^2}$$

Q3. Using multipole expansion, discuss systematic expansion for the potential of any localised charge distribution in powers of $1/r$

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$$V(r) = \frac{1}{4\pi\epsilon_0}\int \frac{1}{\mathscr{r}}\rho(r')\,d\tau'$$

Using law of cosines

$$\mathscr{r}^2 = r^2 + (r')^2 - 2rr'\cos\alpha = r^2\left[1 + \left(\frac{r'}{r}\right)^2 - 2\left(\frac{r'}{r}\right)\cos\alpha\right]$$

$$\mathscr{r} = r\sqrt{1+\epsilon}$$

$$\epsilon \equiv \left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right)$$

For points outside charge distribution, ϵ is much less than 1

$$\frac{1}{\mathscr{r}} = \frac{1}{r}(1+\epsilon)^{-1/2} = \frac{1}{r}\left(1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \frac{5}{16}\epsilon^2 + \cdots\right)$$

$$\frac{1}{\mathscr{r}} = \frac{1}{r}\left[1 - \frac{1}{2}\left(\frac{r'}{r}\right)\left(\frac{r'}{r} - 2\cos\alpha\right) + \frac{3}{8}\left(\frac{r'}{r}\right)^2\left(\frac{r'}{r} - 2\cos\alpha\right)^2 - \frac{5}{16}\left(\frac{r'}{r}\right)^3\left(\frac{r'}{r} - 2\cos\alpha\right)^3 + \cdots\right]$$

$$= \frac{1}{r}\left[1 + \left(\frac{r'}{r}\right)(\cos\alpha) + \left(\frac{r'}{r}\right)^2\left(\frac{3\cos^2\alpha - 1}{2}\right) + \left(\frac{r'}{r}\right)^3\left(\frac{5\cos^3\alpha - 3\cos\alpha}{2}\right) + \cdots\right]$$

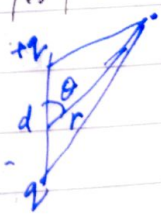
Legendre Polynomials

$$\frac{1}{\mathscr{r}} = \frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\alpha)$$

$$V(r) = \frac{1}{4\pi\epsilon_0}\sum_{n=0}^{\infty}\frac{1}{r^{(n+1)}}\int (r')^n P_n(\cos\alpha)\,\rho(r')\,d\tau'$$

Example 3.10

Using multipole expansion find approx potential for from dipole



$$V(r) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_+} - \frac{q}{r_-}\right)$$

$$r_\pm^2 = r^2 + (d/2)^2 \mp rd\cos\theta = r^2\left(1 \mp \frac{d}{r}\cos\theta + \frac{d^2}{4r^2}\right)$$

$r \gg d$

$$\frac{1}{r_\pm} \cong \frac{1}{r}\left(1 \mp \frac{d}{r}\cos\theta\right)^{-1/2} \cong \frac{1}{r}\left(1 \pm \frac{d}{2r}\cos\theta\right)$$

$$\frac{1}{r_+} - \frac{1}{r_-} \cong \frac{d}{r^2}\cos\theta$$

$$V(r) \cong \frac{1}{4\pi\epsilon_0}\frac{qd\cos\theta}{r^2}$$

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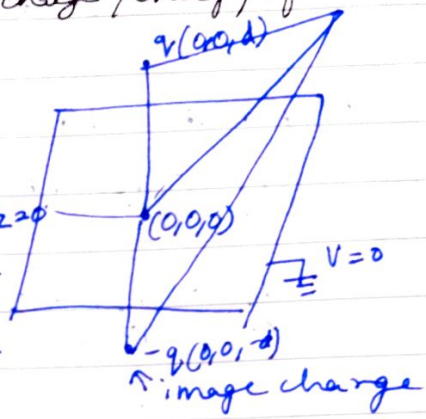
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Q86 A charge q is at a distance d above an infinite grounded conducting plane. Find

(i) Potential

(ii) Electric field in the region above the plan

Also find force b/w plane & the charge, energy of this config.



(i) $V = 0$ at $z = 0$

(ii) $V \to 0$ if $r^2 = x^2 + y^2 + z^2 >> d^2$

$$V = \frac{q}{4\pi\epsilon_0\sqrt{x^2+y^2+(z-d)^2}} - \frac{q}{4\pi\epsilon_0\sqrt{x^2+y^2+(z+d)^2}}$$

This is the potential of point charge & image of point charge

We replace the prob of a point charge into the prob of 2 pt. charges → one q at $(0,0,+d)$ & $-q$ at $(0,0,-d)$

electric field $= \vec{E} = -\vec{\nabla}V$

$$= -\partial V/\partial z\ \hat{z}$$

$\vec{E}$ will be in z dir^n only bcoz x & y axis are infinite

$$\vec{E} = -\partial/\partial z\left[\frac{q}{4\pi\epsilon_0\sqrt{x^2+y^2+(z-d)^2}} - \frac{q}{4\pi\epsilon_0\sqrt{x^2+y^2+(z+d)^2}}\right]\hat{z}$$

$$= \frac{-q}{4\pi\epsilon_0}\left[-\frac{1}{2}\left(x^2+y^2+(z-d)^2\right)^{-3/2}(2z-2d) + \frac{1}{2}\left(x^2+y^2+(z+d)^2\right)^{-3/2}\right]$$

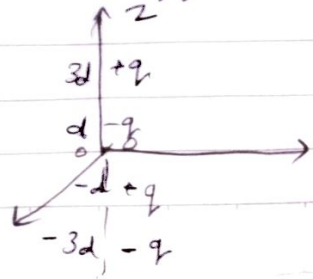
$$\vec{E} = \frac{+q}{4\pi\epsilon_0}\left[\frac{(z-d)}{\left(x^2+y^2+(z-d)^2\right)^{3/2}} - \frac{(z+d)}{\left(x^2+y^2+(z+d)^2\right)^{3/2}}\right]\hat{z}$$

This is elecfield at distance z above above plane

Electric field below plane $\vec{E} = 0$

Q91 Find the force on a charge $+q$ if x-y plane is a grounded conductor. Also find Energy of this configuration

First make the image charge setup



$$\vec{F}_q = \frac{+1}{4\pi\epsilon_0}\frac{q^2}{4d^2}(-\hat{z}) + \frac{1}{4\pi\epsilon_0}\frac{q^2}{16d^2}(\hat{z}) + \frac{1}{4\pi\epsilon_0}\frac{q^2}{3d^2}(-\hat{z})$$

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$$= -\frac{q^2}{4\pi\epsilon_0 d^2}\left[\frac{1}{4} - \frac{1}{16} + \frac{1}{36}\right]\hat{z}$$

$$= -\frac{q^2}{4\pi\epsilon_0 d^2}\left[\frac{36-9+4}{144}\right]\hat{z}$$

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{q^2}{d^2}\left(\frac{-31}{144}\right)\hat{z}$$

Energy of the configuration

$$W_1 = -\int_{\infty}^{3d} F \cdot dr$$

$$= -\int_{\infty}^{3d} \frac{1}{4\pi\epsilon_0}\frac{q^2}{z^2}\left(\frac{-31}{144}\right)dz$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q^2 31}{144}\left[\frac{-1}{z}\right]_{\infty}^{3d}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{31q^2}{144}\left(\frac{-1}{3d}\right)$$

$$\boxed{W_1 = \frac{-31q^2}{4\pi\epsilon_0(432d)}}$$

Work done to take charge $-q$ from ∞ to,

$$F = \frac{1}{4\pi\epsilon_0}\frac{q^2(-\hat{z})}{4d^2}$$

$$W_2 = -\int_{\infty}^{d} F \cdot dr$$

$$= -\int_{\infty}^{d} -\frac{1}{4\pi\epsilon_0}\frac{q^2}{4z^2}dz$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q^2}{4}\left[\frac{-1}{z}\right]_{\infty}^{d} \Rightarrow \boxed{W_2 = \frac{-q^2}{4\pi\epsilon_0(4d)}}$$

Total Energy of System is $W = W_1 + W_2$

$$W = \frac{-31q^2}{4\pi\epsilon_0(432d)} - \frac{q^2}{4\pi\epsilon_0(4d)}$$

$$W = \frac{-q^2}{4\pi\epsilon_0 d}\left(\frac{31}{432} + \frac{1}{4}\right)$$

$$W = \frac{-q^2}{4\pi\epsilon_0}\left(\frac{139}{432d}\right)$$

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Chapter 4: Electric Field in Matter

Theory

Q1. Write the diff between conductors, insulators & dielectrics

	Conductor	Insulators	Dielectrics
1	Large no. of free charges	Very few/no free charges	Very few/no free charges
2	Charges can move freely	Charges cannot move freely	Charges cannot move freely but bound charges can slightly shift
3	Elec field inside = 0	Can be non-zero	Can be non zero
4	Polarisation - not normally described by bound charge	May be very small	Significant
5	In external field - free charge redistribute	Little charge movement	Molecules/atoms become polarized
6	Copper, aluminium	Glass, rubber, plastic	Glass, mica, ceramic

Q2. Explain electric polarisation.

When a dielectric is placed in an external field, the +ve and -ve charges within its molecules experience forces in opposite dir^n. As a result, the centers of +ve & -ve charges become slightly separated. This produces electric dipole inside the material. This phenomena is called electric polarisation

$$P = \varepsilon_0 \chi_e E$$

Q3. Explain bound charges using the expression of the potential of a polarised object

These are charges bound or restricted to specific locations within the material. Bound charges do not contribute to the flow of electric current. However the presence and distribution can influence the overall elec field and electric polarization within materials

$$\rho_{bound} = -\nabla \cdot P$$

$$\text{Potential of dipole} = V(r) = \frac{1}{4\pi\varepsilon_0} \int_V \frac{P(r')}{s^2} \hat{s}\, d\tau'$$

$$\nabla\left(\frac{1}{s}\right) = \frac{\hat{s}}{s^2}$$

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$$V = \frac{1}{4\pi\epsilon_0}\int_v P\cdot\nabla'\left(\frac{1}{r}\right)d\tau$$

$$V = \frac{1}{4\pi\epsilon_0}\left[\int_v \nabla'\cdot\left(\frac{P}{r}\right)d\tau' - \int_v \frac{1}{r}(\nabla'\cdot P)\,d\tau\right]$$

$$V = \frac{1}{4\pi\epsilon_0}\oint_S \frac{1}{r}P\cdot da' - \frac{1}{4\pi\epsilon_0}\int_v \frac{1}{r}(\nabla'\cdot P)\,d\tau'$$

$$\sigma_b \equiv P\cdot\hat{n}$$

$$\rho_b \equiv -\nabla\cdot P$$

Q4. Explain Gauss's law in presence of Dielectrics.

Within dielectric, the total charge density can be written:

$$\rho = \rho_b + \rho_f$$

Gauss law $\epsilon_0\nabla\cdot E = \rho = \rho_b + \rho_f = -\nabla\cdot P + \rho_f$

where E is now the total field, not just that portion generated by polarization

$$\nabla\cdot(\epsilon_0 E + P) = \rho_f$$

electric displacement → $D \equiv \epsilon_0 E + P$

$$\nabla\cdot D = \rho_f$$

$$\oint D\cdot da = Q_{enc}$$

Q5. Explain electric susceptibility, permittivity, and dielectric constant.

for a linear dielectric, $\boxed{\vec{P} = \epsilon_0\chi_e\vec{E}}$ where χ_e is electric susceptibility

It measures how easily the material becomes polarised in elec field

$$\vec{D} = \epsilon_0\vec{E} + \vec{P} \rightarrow \vec{D} = \epsilon_0\vec{E} + \epsilon_0\chi_e\vec{E} = \epsilon_0(1+\chi_e)E$$

$$\vec{D} = \epsilon\vec{E}$$

$$\epsilon = \epsilon_0(1+\chi_e)$$

ϵ is called permittivity of material. For vacuum $\epsilon = \epsilon_0$

Dielectric constant is usually defined as the relative permittivity

$$\epsilon_r = \epsilon/\epsilon_0$$

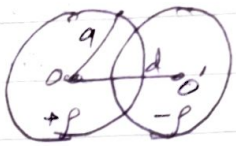
$$\epsilon_r = 1 + \chi_e$$

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Q109 A primitive model of an atom consist of a point charge $+q$ surrounded by an uniformly charged spherical cloud $(-q)$ of radius a. Calculate the atomic polarisability of the atom



$$\vec{E} = \frac{\rho \vec{d}}{3\epsilon_0}$$

$$\vec{E} = \frac{3q\vec{d}}{4\pi\epsilon_0 a^3} \qquad \left(q = \rho V,\ \rho = \frac{q}{V} = \frac{q}{\frac{4}{3}\pi a^3}\right)$$

$$\boxed{E = E_e = \frac{qd}{4\pi\epsilon_0 a^3}} \quad — (1)$$

Dipole moment $p = qd$

$$\boxed{p = 4\pi\epsilon_0 a^3 E} \quad — (2)$$

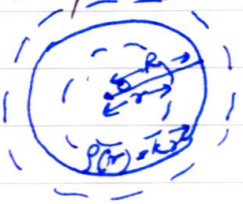
$$p = \alpha E \quad — (3)$$

On comparing we get $\alpha = 4\pi\epsilon_0 a^3$ or $\alpha = 3\epsilon_0 V$

Q112 A shape of radius R carries a polarisation $P(r) = k \cdot \vec{r}$ where k is a constant & $\vec{r}$ is the vector from centre

(a) Calculate bound charges σ_b & ρ_b

(b) Calculate the elec field inside & outside the sphere



(a) $\sigma_b = \vec{P} \cdot \hat{n}$

$= k\vec{r} \cdot \hat{n} = kr\hat{r} \cdot \hat{r}$

$= kr = KR$ [at surface $r = R$

$\rho_b = -\vec{\nabla} \cdot \vec{P}$

$= -\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 k \cdot r)$

$= -\frac{k}{r^2} 3r^2 = -3k$

(b) Inside $q_{enc} = \int \rho_b \, d\tau = -3k \cdot \frac{4\pi}{3} r^3$. (q_{enc} due to volume only)

$$\boxed{q_{enc} = -4\pi k r^3}$$

By Gauss law, $\oint E \cdot ds = q_{enc}/\epsilon_0$

$E \cdot 4\pi r^2 = -\frac{4\pi}{\epsilon_0} k r^3$

$$\boxed{E_{in} = \frac{-kr}{\epsilon_0}\hat{r}}$$

Outside the sphere

$q_b(\sigma_b) = KR\, 4\pi R^2$

$q_b(\sigma_b) = 4\pi R^3 K$

$q(\rho_b) = -4\pi K R^3$

Total $q = 4\pi K R^3 - 4\pi K R^3$

$q = 0$

$E_{out} = 0$

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Q114 A long straight wire carrying a line charge λ is surrounded by rubber insulation upto radius a. Find electric displacement D & Electric field inside & outside the rubber insulation



Inside.- $\oint D \cdot ds = q_{enc}$

$q_{enc} = \lambda l$

$D = 2\pi r l = \lambda l$

$D = \frac{\lambda}{2\pi s}\hat{s}$

$D = \epsilon_0 E + P$ also $\vec{D} = \epsilon \vec{E}$

$\epsilon E = \frac{\lambda}{2\pi s}\hat{s}$

$E_{in} = \frac{D}{\epsilon_0 \epsilon_r} = \frac{\lambda}{2\pi \epsilon_0 \epsilon_r s}\hat{s}$

Outside q free enclosed $q_{enc} = \lambda l$

$D \cdot 2\pi s l = \lambda l$

$$\boxed{D = \frac{\lambda}{2\pi s}\hat{s}}$$

In free space $\epsilon_r = 1$

so $$\boxed{E_{out} = \frac{\lambda}{2\pi \epsilon_0 s}\hat{s}}$$

Q118 Suppose the field inside a large piece of dielectric is E_0 such that electric displacement is D_0 s.t $\vec{D_0} = \epsilon_0 \vec{E} + \vec{P}$

(a) Now a spherical cavity is hollowed out of the material. Find electric field & displacement vector inside the cavity

(b) To the same, for a long needle shaped cavity running parallel to $\vec{P}$

(c) To the same for a thin wavoup shaped cavity $\perp$ to $\vec{P}$

Inside Spherical Cavity

Polarisation, $P = 0$ $+P - P = 0$

bcoz of $+P$ the elec field will be E_0

& bcoz of $-P$ the elec field will be $-\vec{P}/3\epsilon_0$

$$\vec{E} = \frac{-\vec{P}}{3\epsilon_0}$$

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So total elec field in spherical cavity $= \vec{E_0} + \frac{\vec{P}}{3\epsilon_0}$

E_0 is external applied elec field then due to external field, internal field will be produced $P/3\epsilon_0$

$$\text{Total} = E_0 - \frac{P}{3\epsilon_0} < E_0$$

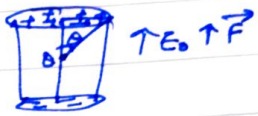
$$\vec{D_0} = \epsilon_0 \vec{E_0} + \vec{P}$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E_0} + \frac{\vec{P}}{3}$$

$$\vec{D} = D_0 - \vec{P} + \frac{\vec{P}}{3} \Rightarrow \boxed{\vec{D} = \vec{D_0} - \frac{2\vec{P}}{3}}$$

Needle shaped Cavity



$$\boxed{\vec{E} = \vec{E_0} - \frac{\vec{P}}{\epsilon_0}(1-\cos\theta)}$$

for needle shaped cavity $\boxed{\theta = 0^\circ}$

$$\vec{E} = \vec{E_0} - \frac{\vec{P}}{\epsilon_0}(1-1)$$

$$\vec{E} = \vec{E_0}$$

$$\vec{D} = \epsilon_0 \vec{E} = \epsilon_0 \vec{E_0} - P + P$$

$$\boxed{\vec{D} = \vec{D_0} - \vec{P}}$$

Unpour Shape $\boxed{\theta = \pi/2}$



$$\vec{E} = \vec{E_0} + \frac{\vec{P}}{\epsilon_0}$$

$$P = 0 \Rightarrow \underset{\epsilon_0}{+P} \underset{-P/\epsilon_0}{-P} = 0 \Big\} \rightarrow \text{by superposition principle}$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$= \epsilon_0 \vec{E_0} + P$$

$$\vec{D} = \vec{D_0}$$

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Chapter 5 : Magnetostatics

Theory

Q1. Discuss the expression of magnetic force on (i) a moving charge and on (ii) a current carrying wire

(i) A charged particle moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences a magnetic force

$$\boxed{\vec{F} = q(\vec{v} \times \vec{B})}$$

(ii) Consider a wire carrying current I in a magnetic field $\vec{B}$

The magnetic force on a small length $d\vec{l}$ is $\boxed{d\vec{F} = I(d\vec{l} \times \vec{B})}$

For a straight wire of length $\vec{L}$: $\boxed{\vec{F} = I(\vec{L} \times \vec{B})}$

Q2. Explain Biot-Savart Law

Biot-Savart law gives the magnetic field produced at a point due to a small current element.

$$B(r) = \frac{\mu_0}{4\pi} \int \frac{I \times \hat{r}}{r^2} dl' = \frac{\mu_0}{4\pi} I \int \frac{dl' \times \hat{r}}{r^2}$$

Biot-Savart law is the magnetic equivalent of the electrostatic field calculation using Coulomb's law.

Q3 Discuss the significance of the divergence of the magnetic field.

Maxwell's equation for magnetostatics $\nabla \cdot B = 0$

The magnetic field has zero divergence everywhere

Magnetic field lines do not begin or end at any point. They always form closed loops. Magnetic monopoles have not been observed.

Q4. Discuss the significance of curl of magnetic field (ampere's law)

Ampere law states that circulation of magnetic field around a closed path is proportional to current enclosed by that path.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

The curl measures the circulation or rotational tendency of a vector fi

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

Electric ≠ current acts as a source of circulation of magnetic field

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Q5. Explain magnetic vector potential and compare it with scalar electrostatic potential

$\nabla \cdot B = 0$

$B = \nabla \times A$ ← magnetic vector potential

In Magnetic vector potential, Add to A any function whose curl vanishes (which is to say, the gradient of any scalar)

This will not affect B

$A' = A + \nabla\lambda$ Gauge Transform

$\nabla \times B = \mu_0 J$

$\nabla \times B = \nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A = \mu_0 J$

$\nabla^2 A = -\mu_0 J$

$A(r) = \frac{\mu_0}{4\pi} \int \frac{J(r')}{\ell} d\tau'$

In electrostatics, $\vec{E} = -\nabla V$; derived from Poisson's equation

where V is a scalar potential

The electrostatic potential is determined by its gradient

Q6. Describe Boundary condition for Magnetostatics

Magnetic field is discontinuous at a surface current

$B^{+}_{above} = B^{+}_{below}$

$B^{\parallel}_{above} - B^{\parallel}_{below} = \mu_0 K$

Vector potential is continuous across any boundary

$A_{above} = A_{below}$.

Q7. Describe the multipole expansion of the vector potential

Approximate formula for vector potential of a localized current distribution which is valid at distant points

$A(r) = \frac{\mu_0 I}{4\pi} \oint \frac{1}{\ell} dl'$

monopole term goes like $1/r$

dipole term goes like $1/r^2$

quadrupole term goes like $1/r^3$

Magnetic monopole term is always zero

Magnetic dipole term can be written as

$A_{dip}(r) = \frac{\mu_0}{4\pi} \frac{m \times \hat{r}}{r^2}$

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Q An e^- enters in a uniform mag field with magnitude 0.3 tesla at an angle of 45° wrt to mag field
Determine the radius r & pitch p of the electron's helical path. Assuming its speed is 2×10^6 m/s



$\theta = 45°$, $v = 2\times10^6$ m/sec

$v_\perp = v\sin 45° = 2\times10^6 \times 1/\sqrt{2} = \sqrt{2}\times10^6$

$v_{||} = v\cos 45° = 2\times10^6 \times 1/\sqrt{2} = \sqrt{2}\times10^6$

$v_{||} = v_\perp = \sqrt{2}\times10^6$ m/sec

$v_\perp$ is responsible for circular motion

$v_{||}$ is responsible for forward motion

$$qv_\perp \times B = mv_\perp^2/r$$

$$r = mv_\perp/qB = \frac{9.1\times10^{-31}\times\sqrt{2}\times10^{16}}{1.6\times10^{-19}\times0.3}$$

$$r = 26.81.087\times10^{-6}$$

$$\boxed{r = 26.8\ \mu m}$$

$$\Delta t = 2\pi r/v_\perp$$

pitch $p = v_{||}\times\Delta t$

$= v_{||}\times 2\pi r/v_\perp$

$p = 2\pi r = 2\times3.14\times26.8 = 168.304$

$$\boxed{p = 169\ \mu m}$$

Q a) If Current I is uniformly distributed over a wire of circular cross section with radius a find the volume of current density $\vec{J}$

b) If current in the wire is proportional to the distance from the axis $J = Ks$. Find the total current. K is any current

(a) $J = ?$ $\quad \vec{J} = I/a_\perp = I/\pi a^2$

(b) $J = Ks$

$I = ?$ $\quad J = I/a_\perp$

This current density is Non Uniform so

$$I = \int J\cdot da_\perp$$

$$\int_0^{2\pi}\int_0^a ks\cdot s\, ds\, dp$$

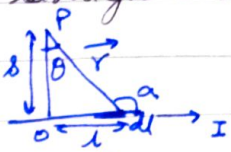
$$= k\left(\frac{s^3}{3}\right)_0^a 2\pi$$

$$I = \frac{2\pi k a^3}{3}$$

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Q Find the mag field at a distance s from a straight wire carrying a steady current I



Take a small current dl.

dir^n of element dl will be in the dir^n of flow of current

$$dB = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \vec{r}}{r^3}$$

Complete mag field for wire

$$B = \int dB = \frac{\mu_0}{4\pi} I \int \frac{d\vec{l} \times \vec{r}}{r^3}$$

$$= \frac{\mu_0 I}{4\pi} \int \frac{dl\, r \sin\alpha}{r^3}$$

$$\frac{\mu_0 I}{4\pi} \int \frac{dl \sin\alpha}{r^2}$$

$$90 + 180 - \alpha + \theta = 180$$

$$\boxed{\alpha = 90 + \theta}$$

$$\therefore \sin\alpha = \sin(90+\theta) = \cos\theta$$

$$B = \frac{\mu_0 I}{4\pi} \int \frac{\cos^2\theta}{s^2} \cos\theta\, dl$$

$\tan\theta = l/s$
$l = s\tan\theta$
$dl = s\sec^2\theta\, d\theta$

$$= \frac{\mu_0 I}{4\pi} \int \frac{\cos^2\theta}{s^2} \cos\theta\, s \sec^2\theta\, d\theta$$

$$= \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos\theta\, d\theta$$

$$\boxed{B = \frac{\mu_0 I}{4\pi s} [\sin\theta_2 - \sin\theta_1]}$$ finite wire

For infinitely long wire

then $\theta_1 = -90^\circ$, $\theta_2 = +90^\circ$

$$B = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

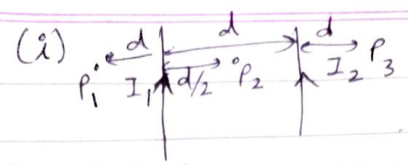
$\hat{\phi}$ → circumfrontial vector

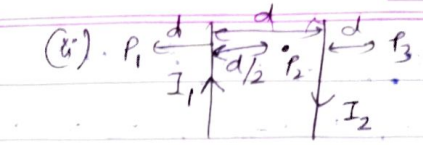
Q Find the force per unit length in 2 parallel wire arrangement as shown in figure.

Also find the mag field at point P_1, P_2 & P_3

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(i) 

(ii) 

$B = \frac{\mu_0 I}{2\pi s}\hat{\phi}$ force on wire 2 will be due to mag. field of wire 1

$\vec{F}_2 = I_2 \int d\vec{l}_2 \times \vec{B}_1$ (force on wire 2)

force per unit length is $f_2 = \vec{I}_2 \times \vec{B}_1$

$\vec{B}_1$ → mag. field on wire 1 at the position of 2

$\vec{B}_1 = \mu_0 I_1 / 2\pi d$ (into the page)

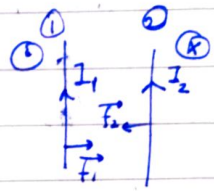
$$\boxed{\vec{F}_2 = \frac{\mu_0 I_1 I_2}{2\pi d}}$$ (dir^n → to the left)

force / unit length $\propto I_1 \& I_2$
$\propto 1/d$

force on wire 1,

$\vec{F}_1 = I_1 \int d\vec{l}_1 \times \vec{B}_2$

$\vec{f}_1 = \vec{I}_1 \times \vec{B}_2$



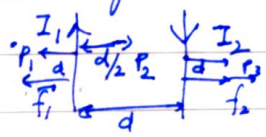
$B_2 = \mu_0 I_2 / 2\pi d$ (out of page)

$$\boxed{\vec{f}_1 = \mu_0 I_1 I_2 / 2\pi d}$$ (to the right)

(ii) Magnitude of force per unit length is same. Only dir^n will be diff

dir^n of mag field on wire (2) due to (1) is ⊗

dir^n of mag field on wire (1) due to (2) is ⊗



At P_1, mag field $\vec{B} = \vec{B}_1 + \vec{B}_2$

$\vec{B}_1 \rightarrow \odot$, $\vec{B}_2 \rightarrow \otimes$

$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi d}$ (out of page)

$\vec{B}_2 = \frac{\mu_0 I_2}{2\pi (2d)}$ (into the page)

$\vec{B} = \vec{B}_1 + \vec{B}_2 \Rightarrow \vec{B} = \frac{\mu_0}{2\pi d}\left[I_1 - \frac{I_2}{2}\right]$

At P_2, $\vec{B}_1 \rightarrow \otimes$ ~~$\vec{B}_2 \rightarrow \otimes$~~

$B = \frac{\mu_0}{2\pi}\left[\frac{1}{d/2}(I_1 + I_2)\right]$

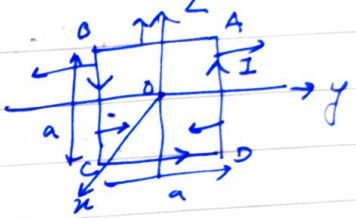
$B = \frac{\mu_0}{2\pi}\frac{2}{d}(I_1 + I_1)$

At P_3, $B_1 \rightarrow \otimes$, $B_2 \rightarrow \odot$

$$B_1 = \frac{\mu_0 I_1}{2\pi(2d)}, \quad B_2 = \frac{\mu_0 I_2}{2\pi(d)}$$

$$B(P_3) = \frac{\mu_0}{2\pi d}\left[\frac{-I_1}{2} + I_2\right]$$

Q) A Mag. field in some region is given by $\vec{B} = kz\hat{x}$ where k is constant. Find the force on a square loop of sides a lying in the y-z plane & centered at the origin if it carries a current I in the ~~same~~ counter clockwise dir^n when you look down the x-axis



$$F = \int I\,d\vec{l} \times \vec{B}$$
$$= I \int d\vec{l} \times \vec{B}$$

For AB wire, mag field $B = \frac{ka}{2}\hat{x}$

$$\vec{F}_{AB} = \frac{Ika^2}{2}\hat{z}$$

On wire CD, Mag field $B = ka/2\ (-\hat{x})$

$$\vec{F}_{CD} = Ika^2/2 \cdot \hat{z}$$

On AD upper half, Dir^n of $\vec{B} \Rightarrow \hat{x}$

on BC upper half, Dir^n of $\vec{B} \rightarrow +\hat{x}$

on AD lower half Dir^n of $\vec{B} \rightarrow -\hat{x}$

on BC lower half Dir^n of $\vec{B} \rightarrow -\hat{x}$

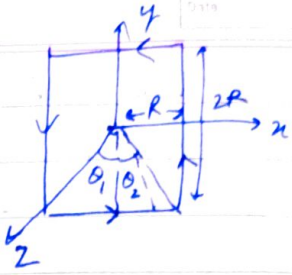
$$\vec{F}_{DA} = \vec{F}_{BC} = 0$$

$$\vec{F} = \left(\frac{Ika^2}{2} + \frac{Ika^2}{2}\right)\hat{z} \Rightarrow F = Ika^2$$

Q) (a) Find the magnetic field at the centre of a square loop which carries steady current I. R be the distance from Centre to sides

(b) Find the field at the centre of a N side polygon carrying a steady current I, for all the wires dir^n of B is outside

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Mag field at centre will be 4 times of mag field due to one wire.

Mag field of for a finite wire

$$B = \frac{\mu_0 I}{4\pi s}\left[\sin\theta_2 - \sin\theta_1\right]$$

for lower wire $B = \frac{\mu_0 I}{4\pi R}\left[\sin 45 - (-\sin 45^\circ)\right]$

$$= \frac{\mu_0 I}{4\pi R} \times 2 \times \sin 45 = \frac{\mu_0 I}{4\pi R}\, 2 \times \frac{1}{2}$$

$$\vec{B} = \frac{\mu_0 I \sqrt{2}}{4\pi R}\hat{z}$$

∴ for all the sides i.e. for square loop

B = 4 times of (B for one wire)

$$\boxed{\vec{B} = \frac{\sqrt{2}\,\mu_0 I}{\pi R}\hat{z}}$$

(b) general formula

mag field for n-side polygon is

$$B = \frac{n\mu_0 I}{2\pi R}\sin\left(\frac{\pi}{n}\right)$$

R = ⊥ distance from midpt of wire

Now check it for Square loop

$$\vec{B} = \frac{4\mu_0 I}{2\pi R}\sin\left(\frac{\pi}{4}\right) = \frac{2\mu_0 I}{\pi R}\,\frac{1}{\sqrt{2}}$$

$$\vec{B} = \frac{\sqrt{2}\,\mu_0 I}{\pi R}\hat{z}$$

Circle is ∞ sides polygon $n \to \infty$, $\theta \to 0$

$$B = \frac{n\mu_0 I}{2\pi R} \cdot \frac{\pi}{n}$$

$$\boxed{B = \frac{\mu_0 I}{2R}}$$

Q Find the magnetic field at distance s from a long straight wire carrying a steady current I

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Sketch a Ampeical loop set points should lie on the circumference of loop



$I_{enc} = I$

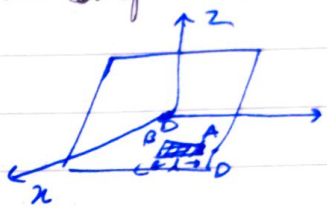
Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

$B \cdot 2\pi s = \mu_0 I$

$$\boxed{\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}}$$

1. Find the mag field of an infinite uniform surface current $\vec{K} = K\hat{x}$ flowing over xy plane

Ampeical loop should be choosen s.t. current must pass through it.



surface is very thin

$\oint \vec{B} \cdot d\vec{l} = \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA}$

If we put too many wires then together they make a surface

$\int_{AB} \vec{B} \cdot d\vec{l} = B \cdot l \cos 0 = Bl$

$\int_{BC} \vec{B} \cdot d\vec{l} = B \cdot l \cos 90 = 0$

$\int_{CD} \vec{B} \cdot d\vec{l} = B \cdot l \cos 90 = Bl$

$\int_{DA} \vec{B} \cdot d\vec{l} = B \cdot l \cos 90 = 0$

So $Bl + 0 + Bl + 0 = \mu_0 I_{enc}$

$K = I/l \Rightarrow I/l = K \Rightarrow I = Kl$

So $2Bl = \mu_0 Kl \Rightarrow B = \mu_0 K/2$

$$\boxed{\begin{aligned} \vec{B} &= \mu_0 K/2\,(-\hat{y}), \quad z>0 \quad \text{(upper surface)} \\ \vec{B} &= \mu_0 K/2\,(+\hat{y}), \quad z<0 \quad \text{(lower surface)} \end{aligned}}$$

This mag field is independent of distance

replace $\mu_0 \to 1/\epsilon_0$ & $K \to \sigma$

$$\vec{E} = \frac{\sigma}{2\epsilon_0}(+\hat{z}), \quad z>0$$

$$\vec{E} = \frac{\sigma}{2\epsilon_0}(-\hat{z}), \quad z<0$$

So $\vec{E}$ & $\vec{B}$ are equivalent

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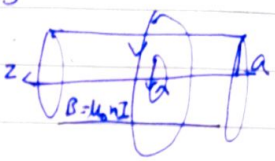
Q. Find the mag vector potential of an ∞ solenoid with n turns per unit length, radius R & current I :

Here current is extended to ∞ i.e. Not localised so can't use above formula to find $\vec{A}$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Mag flux passing through any surface S

$$\phi_m = \int \vec{B} \cdot d\vec{S} = \int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{S}$$
$$= \oint \vec{A} \cdot d\vec{l}$$



Mag field inside the solenoid

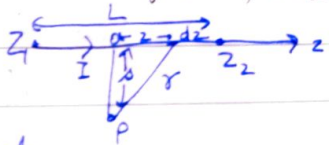
$$B = \mu_0 n I$$
$$\phi_m = B \pi s^2$$
$$= \mu_0 n I \pi s^2 = \oint \vec{A} \cdot d\vec{l}$$
$$\mu_0 n I \pi s^2 = A \cdot 2\pi s$$
$$\Rightarrow \boxed{\vec{A}_{in} = \frac{\mu_0 n I s}{2} \hat{\phi}}$$

Outside

$$\phi_m = B \cdot \pi a^2$$
$$= \mu_0 n I \cdot \pi a^2 = \oint \vec{A} \cdot d\vec{l}$$
$$= A_{out} \cdot 2\pi s$$
$$A_{out} = \frac{\mu_0 n I a^2}{2s} \hat{\phi}$$

Q Find the mag vector potential of a finite segment of straight wire carrying current I. Put the wire along z axis from z_1 to z_2

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{I\, dl}{r}$$



Take a small element, its length is dz which is at a distance z from mid pt. of wire O

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{z_1}^{z_2} \frac{I\, dz}{(s^2+z^2)^{1/2}} = \frac{\mu_0}{4\pi} \int_{-L/2}^{L/2} \frac{I\, dz}{(s^2+z^2)^{1/2}}$$

$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln \left[z + \sqrt{z^2+s^2} \right]_{-L/2}^{L/2}$$

$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln \left[\frac{L/2 + \sqrt{\frac{L^2}{4} + s^2}}{-L/2 + \sqrt{\frac{L^2}{4} + s^2}} \right]$$

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$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln\left[\frac{L/2 + (L^2/4 + \delta^2)^{1/2}}{-L/2 + (L^2/4 + \delta^2)^{1/2}}\right]$$

If wire is very very long $L \gg \delta$

$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln\left[\frac{L/2\left[1 + \left(1 + \frac{4\delta^2}{L^2}\right)^{1/2}\right]}{L/2\left[-1 + \left(1 + \frac{4\delta^2}{L^2}\right)^{1/2}\right]}\right]$$

$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln\left[\frac{1 + 1 + \frac{2\delta^2}{L^2}}{-1 + 1 + \frac{2\delta^2}{L^2}}\right] = \frac{\mu_0 I}{4\pi} \ln\left[\frac{2 + \frac{2\delta^2}{L^2}}{\frac{2\delta^2}{L^2}}\right]$$

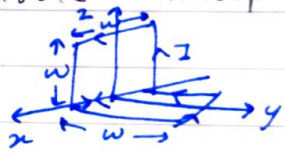
$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln\left[\frac{L^2}{\delta^2} + 1\right]$$

$L \gg \delta$ then $\frac{L}{\delta} \gg 1$ & $L^2/\delta^2 \gg\gg 1$ so neglect 1 as compared to L^2/δ^2 so

$$\vec{A} = \frac{\mu_0 I}{4\pi} \ln\left(\frac{L}{\delta}\right)^2$$

$$\boxed{\vec{A} = \frac{\mu_0 I}{2\pi} \ln\left(\frac{L}{\delta}\right)\hat{z}}$$

Q Find the mag dipole moment of a bookend shape loop All sides have length w and current I. Separate both loop & find the mag mom of each loop.



①  This loop is in x-z plane

so dir^n of mag momentum is $\hat{y}$

$$\vec{m_1} = I w^2 \hat{y}$$

Again for 2nd loop, loop is in x-y plane ② 

so dir^n of mag mom is $\hat{z}$

$$\vec{m_2} = I w^2 \hat{z}$$

So total mag momentum, vector sum of $\vec{m_1}$ & $\vec{m_2}$

$$\vec{m} = \vec{m_1} + \vec{m_2} \Rightarrow \boxed{\vec{m} = I w^2 (\hat{y} + \hat{z})}$$

$$\boxed{|\vec{m}| = \sqrt{2}\, I w^2}$$

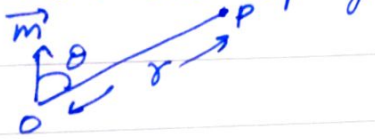
Q A circular loop of wire with radius R lies in the x-y plane centred at the origin & carries current I running counter clockwise as viewed from z axis

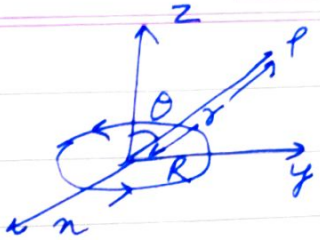
(a) find mag dipole moment

(b) Approximate mag field at large distance

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We can replace the whole loop by a single m





Mag. dipole moment

$$\vec{m} = I\vec{a}$$

$$\boxed{\vec{m} = I\pi R^2 \hat{z}}$$

Mag. field at point P

$$\vec{B} = \frac{\mu_0 m}{4\pi r^3}\left[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right)$$

$$\boxed{\vec{B} = \frac{\mu_0 I}{4\pi r^3}\,\pi R^3\left[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right]}$$

magnetic vector pot^n at the axis of loop, $\theta = 0$

$$\vec{A} = \frac{\mu_0}{4\pi}\,\frac{m\sin\theta}{r^2}\,\hat{\phi}$$

$$\boxed{\vec{A}_{axis} = 0}$$